

PANEL: A photovoltaic-specific vision-language model for zero-shot and few-shot PV interpretation tasks in remote sensing

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ABSTRACT

Large-scale photovoltaic (PV) mapping from remote sensing (RS) is crucial for solar energy assessment and infrastructure development. While vision-language models (VLMs) have advanced language-guided image understanding, existing RS CLIP models underperform on PV across changing regions, sensors, and imaging conditions due to the scarcity of PV imagery and aligned prompts in open RS multi-modal datasets. To address this, we curate the first worldwide PV vision-language dataset, comprising over one million image-text pairs spanning 0.1–20 m resolutions, and design aligned flexible binary prompts that explicitly distinguish PV-present from PV-absent imagery. We then build **PANEL** (PV-specific vision-language MODEL) by retraining CLIP using a tailored pipeline featuring extensive augmentation and prompt diversification. For zero-shot inference, we introduce PANEL Surgery to reinforce vision-language alignment and extract PV-relevant priors for PV interpretation tasks. For few-shot adaptation, we propose a plug-in Knowledge Assistance Module (KAM) that injects these priors into diverse baselines via gated convolution and cross-attention fusion. To enable rigorous evaluation, we establish seven PV-specific benchmarks spanning five interpretation tasks and observe that PANEL consistently surpasses representative CLIP models in zero-shot classification (>40% accuracy), semantic localization (>45% mSC), segmentation (>25% IoU), detection (>24% recall), and image generation (>7% Δ_{CLIP} reduction). In few-shot settings, KAM generalizes across multiple architectures with minimal labels, and ablation studies confirm the necessity of vision-language priors and the complementary design of each component. We further introduce a label-efficient sampling strategy leveraging VLM-derived similarity scores, consistently outperforming random selection under tight annotation budgets. Collectively, these results establish PANEL as a unified, PV-specific VLM for robust zero-/few-shot interpretation while reducing annotation costs in real-world RS for solar energy. PANEL and benchmarks are publicly available at <https://huggingface.co/UEXdo>.

1. Introduction

Photovoltaic (PV) systems have rapidly reshaped the global energy landscape and become central to carbon-neutral development, driving a growing need for efficient and scalable monitoring of their distribution and status (Mohanasundaram and Rangaswamy, 2025). Remote sensing (RS) provides timely and large-scale observations across diverse regions, making it an indispensable tool for PV interpretation, especially for global energy planning and asset management (Kruitwagen et al., 2021).

However, robust PV interpretation from RS imagery remains challenging because PV targets exhibit distinctive complexity across spatial, contextual, and spectral dimensions. Spatially, PV panels show substantial intra-class variability, ranging from fragmented and irregular rooftop installations in dense urban areas to massive and structured utility-scale farms in deserts (Stowell et al., 2020). Contextually, they are frequently affected by shadows, occlusions, and specular reflections, which increase inter-class similarity with buildings, roads, and other man-made structures. Spectrally, their reflectance varies with

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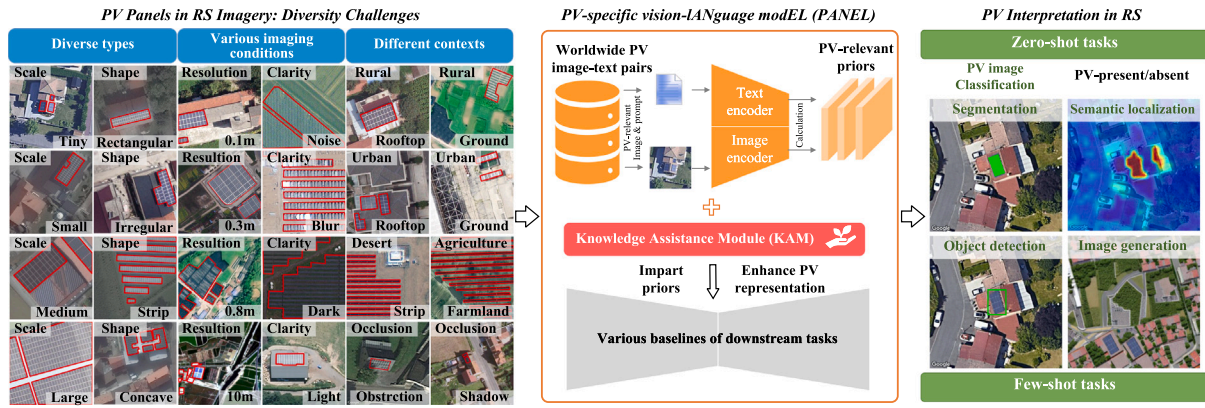


Fig. 1. A PV-specific vision-language model, namely PANEL, using global PV image-text pairs encompassing comprehensive PV scenarios. PANEL inherently supports zero-shot inference for PV-related RS downstream tasks while incorporating a customized Knowledge Assistance Module (KAM) to enhance baselines' performance across few-shot learning tasks.

materials, sensors, and illumination conditions, leading to unstable appearances across regions and resolutions (Chen et al., 2023).

For these complexities, existing PV-oriented RS studies have mainly relied on customized interpretation methods, including architecture design (Guo et al., 2024), task-specific modules (Guo et al., 2023), and image-generation-based data augmentation (Tan et al., 2024, 2026), to improve performance on particular tasks, resolutions, or localized study areas. Although effective in closed-set settings, these methods generally depend on annotated data for training and often generalize poorly across sensors, resolutions, regions, and background conditions. As a result, the PV domain still lacks a highly generalizable foundation model that can support robust zero-shot interpretation or achieve strong performance with only limited fine-tuning data, which is crucial for global-scale monitoring of both distributed and utility-scale PV systems.

In the field of RS, developing foundation models has emerged as a prevalent paradigm to tackle diverse interpretation challenges, primarily evolving along two main directions: vision-only architectures and vision-language models (VLMs). Vision-only models, typically built on masked image modeling (MIM) (Xie et al., 2022) or masked autoencoding (MAE) (He et al., 2022)—such as RingMo (Sun et al., 2022), SatMAE (Cong et al., 2022), and others (Li et al., 2024; F. Wang et al., 2024; D. Wang et al., 2025)—have demonstrated strong representation learning ability. However, these models remain bound by the pretrain-then-finetune paradigm, requiring task-specific heads and costly annotated data. By contrast, VLMs based on the contrastive language-image pre-training (CLIP) paradigm (Radford et al., 2021) align visual representations with textual prompts, offering a robust route to zero-shot transfer without explicit retraining or task-specific heads (Xiang et al., 2025; R. Wang et al., 2025). Additionally, recent studies have actively adapted VLMs to specialized RS tasks through domain-specific retraining and few-shot learning strategies (Cao et al., 2024; Wu et al., 2025). These zero-shot and few-shot capabilities are particularly valuable for global-scale PV monitoring, where collecting exhaustive annotated data for every sensor and region is impractical, and newly constructed solar farms often present extremely limited sample sizes for fine-tuning. Nevertheless, while existing RS VLMs (e.g., RemoteCLIP Liu et al., 2024, GeoRSCLIP Zhang et al., 2024) are developed as powerful generalists, they remain insufficiently exposed to PV-related semantics during pre-training, as PV panels are relatively rare in mainstream RS captioning datasets compared to common RS objects. Therefore, a dedicated VLM for PV interpretation in RS is still missing.

To bridge this gap, we develop a PV-specific VLM tailored for robust PV interpretation across diverse RS scenarios, as illustrated in Fig. 1. We first curate the first worldwide PV vision-language dataset by aggregating open resources, yielding 593,438 PV-present and 447,255

PV-absent samples spanning 0.1–20 m resolutions. To overcome the scarcity of dense captions, we introduce a PV-specific binary prompt strategy that explicitly distinguishes PV-present from PV-absent imagery. We then retrain CLIP end-to-end using a tailored pipeline featuring extensive augmentation, prompt diversification, and balanced sampling, resulting in **PANEL**, a PV-specific vision-language model. To unleash the potential of PANEL across various downstream tasks, we develop a unified adaptation framework, **PANEL Surgery**, designed to extract robust PV-relevant priors from the frozen PANEL. For zero-shot tasks, these priors are directly utilized as prediction results or conditioning guidance, enabling training-free classification, semantic localization, segmentation, object detection, and image generation for PV data augmentation task. For few-shot adaptation, we further propose a plug-in Knowledge Assistance Module (KAM) to inject these PANEL-derived priors into diverse baseline architectures, bridging PANEL's zero-shot semantic generalization with the task-specific learning capacity required by downstream models in data-scarce scenarios. To enable thorough evaluation, we establish PV-specific benchmarks across diverse resolutions and tasks. Comprehensive experiments demonstrate that PANEL consistently outperforms representative CLIP models in zero-shot settings, while KAM proves effective in boosting various baseline architectures under few-shot scenarios. Furthermore, we propose a practical label-efficient sampling strategy based on PANEL similarity scores, offering a cost-effective solution for data annotation under tight budgets.

The main contributions of this paper are as follows:

1. We construct the first worldwide PV vision-language RS dataset and train PANEL, a PV-specific VLM that supports PV image classification, semantic localization, segmentation, detection, and image generation.
2. We propose PANEL Surgery to extract robust PV-relevant priors from PANEL and introduce a plug-in KAM module that injects these priors to enhance few-shot performance across diverse baselines.
3. We establish seven PV benchmarks for all tasks and conduct comprehensive evaluations that show that PANEL and KAM deliver substantial zero/few-shot gains in various tasks.
4. We provide a practical label-efficient sampling strategy that leverages CLIP/PANEL's similarity scores to guide data selection, yielding consistent gains under tight annotation budgets.

2. RS VLMs: Progress and bottlenecks in PV

Driven by the CLIP paradigm, RS VLMs have developed rapidly in recent years. As summarized in Table 1, representative models such as CLIP-RSICD (Arutunian et al., 2021), RS-CLIP (Li et al.,

Table 1
Representative RS vision–language foundation models.

VLMs	Year	Pre-training datasets	Dataset scale	Downstream tasks	Open	Highlight
CLIP (Radford et al., 2021)	2021	Collection from the internet	400M image-text pairs	Image classification; Image-to-text retrieval; Other multi-modal vision-language tasks	✓	The first large-scale VLM
CLIP-RSICD (Arutiunian et al., 2021)	2021	RSICD; Sydney; UCM	63,565 image-text pairs	Image classification; Image-to-image and image-to-text retrieval;	✓	Explore zero-shot CLIP-based RS tools
CLIP-RS (Djoufack Basso, 2022)	2022	RSICD; Sydney; UCM	63,565 image-text pairs	Image-to-image and image-to-text retrieval	✗	Deploy a cross-modal RS retrieval web platform
RS-CLIP (Li et al., 2023)	2023	UCM; WHU-RS19; NWPUR-RESISC45; AID	44,613 images	Scene classification	✓	Introduce a pseudo-labeling and curriculum learning strategy for pre-training
GeoCLIP (Vivanco Cepeda et al., 2023)	2023	MP-16	4.72M image-GPS pairs	Worldwide image geo-localization; Image-to-GPS retrieval	✓	Extend CLIP to geolocation
RemoteCLIP (Liu et al., 2024)	2024	SEG-4; DET-10; RET-3	828,725 image-text pairs	Image classification; Text-to-image and image-to-text retrieval; Object counting	✓	Introduce a unified image-caption format and Remote-Count benchmark
SkyCLIP (Z. Wang et al., 2024)	2024	SkyScript	2.6M image-text pairs	Image classification; Text-to-image and image-to-text retrieval	✓	Connect OpenStreetMap (OSM) with multi-source RS images
GeoRSCLIP (Zhang et al., 2024)	2024	RS5M	5M image-text pairs	Image classification; Semantic localization; Text-to-image and image-to-text retrieval	✓	Develop a domain-adapted VLM
Git-RSCLIP (Liu et al., 2025)	2025	Git-10M	10M image-text pairs	Image classification; Text-to-image retrieval; Image generation; Image editing	✓	Present a diffusion-based RS image generation model, termed Text2Earth

2023), RemoteCLIP (Liu et al., 2024), SkyCLIP (Z. Wang et al., 2024), GeoRSCLIP (Zhang et al., 2024), and Git-RSCLIP (Liu et al., 2025) have extended vision-language pre-training to RS imagery using diverse data sources and supervision strategies. These models have demonstrated promising zero-shot or transferable capability on a range of downstream tasks, including classification, retrieval, semantic localization, generation, and editing, thereby establishing VLMs as an important direction for RS foundation models.

Despite these advances, existing RS VLMs are primarily designed as generalists. Their training data and textual supervision center on common RS scenes, land-cover categories, and generic objects, rather than PV-specific semantics. Consequently, while they exhibit strong vision-language alignment on canonical RS targets (e.g., vehicles, ships, buildings), their capacity to support fine-grained PV interpretation remains unknown, especially given the severe appearance variations caused by scale, sensor, and background diversity. To illustrate this limitation, we visualize the vision-language alignment of representative CLIP models on several PV images using the prompt “a remote sensing imagery with PV panels” (Fig. 2). The resulting activation maps reveal counter-intuitive patterns and a clear mismatch between general RS semantics and PV-specific features: high-activation regions (red) frequently highlight irrelevant background noise or structures, whereas actual PV panels often fall into low-confidence zones (blue). This misalignment is largely attributed to the inherent tendency of standard $q-k-v$ attention to rely on irrelevant background areas rather than concentrating on semantically meaningful objects within the image (Xu et al., 2025; Shi et al., 2026). Such over-reliance on background cues becomes detrimental when PV targets exhibit indistinct spectral characteristics or blurred textures, preventing generalist RS VLMs from achieving the fine-grained spatial alignment necessary for robust PV interpretation.

Furthermore, constructing rich, high-quality PV captions to train a dedicated model poses a significant challenge. Although recent studies

have explored generating text descriptions from existing labels using large language models (Zhang et al., 2024; Liu et al., 2025), applying this to PV imagery requires extensive domain-specific verification to prevent semantic hallucinations and ensure consistency. As a practical trade-off, we investigate a simple yet effective binary prompt formulation that explicitly distinguishes PV-present from PV-absent imagery. While this design may not fully exploit the broader expressive capacity of natural language, it establishes an efficient and robust baseline for developing a PV-specific VLM with meaningful zero-shot capabilities. More importantly, if such single-target semantic prompting proves effective, it could offer a viable pathway for constructing tailored VLMs for other specialized, underrepresented RS targets.

3. Worldwide pv vision-language dataset and benchmarks

In this section, we introduce the construction of the first worldwide PV vision-language dataset, tailored to address the data scarcity in PV-specific multi-modal learning. To ensure transparency and reproducibility, we detail the imagery collection, prompt design, and the rigorous construction of both the pre-training dataset for PANEL and the seven distinct benchmarks for downstream tasks.

3.1. Worldwide imagery collection

We aggregated existing open-source PV-related RS datasets to construct a comprehensive repository covering globally leading PV-adopting regions. As detailed in Table 2, our collection spans spatial resolutions from 0.1 m to 20 m, featuring RGB spectral bands that encompass diverse imaging conditions and complex background contexts.

We utilized all available open-source datasets with spatial resolutions finer than 0.8 m, including both original RS imagery and pre-cropped tiles. These sub-meter datasets primarily capture distributed

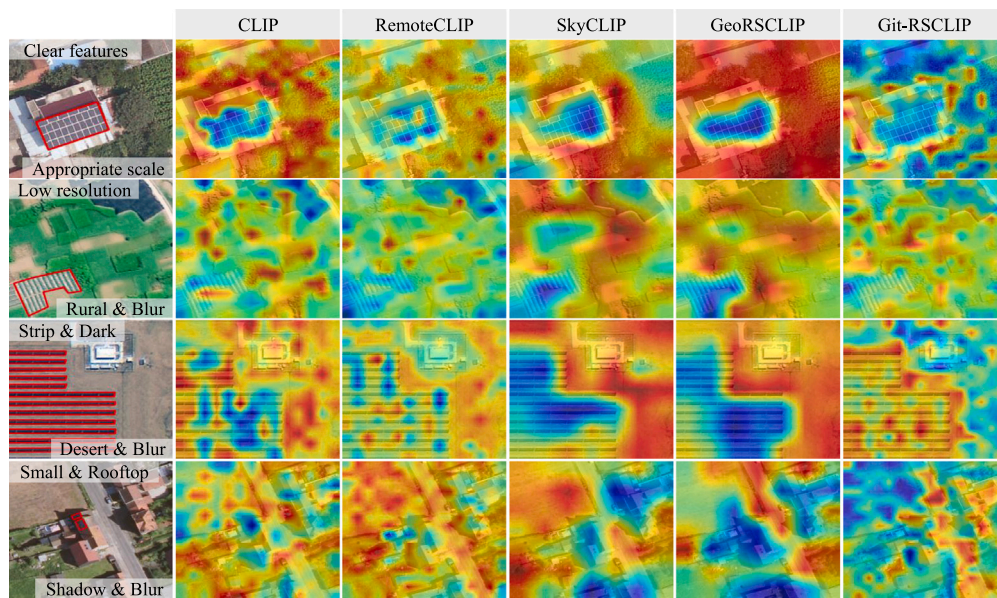


Fig. 2. Visualization of vision-language alignment for PV panels. The alignment is computed by measuring the cosine similarity between text features and spatial features across different CLIP models. The red indicates activated regions with the higher similarity score, while the blue indicate lower score.

PV installations, such as rooftop arrays in urban environments. Conversely, datasets with spatial resolutions coarser than 10 m are typically employed for monitoring large-scale centralized PV plants at national or global scales.

For large-area datasets lacking source imagery, specifically China PV-2022 (Chen et al., 2024) and Global PV-2022 (A. Li et al., 2025), we acquired corresponding imagery via Google Earth Engine (GEE) based on their label files. To ensure consistent input sizes for model training, we generated a circular buffer with a radius of 192 pixels for each PV polygon and downloaded valid imagery within the calculated latitude/longitude bounds. Crucially, to address spatial overlaps between different source datasets, we utilized geospatial indexing to identify duplicate areas. We then merged these overlapping polygons and performed manual verification to refine the boundaries, ensuring a high-quality, non-redundant dataset.

Fig. 3 illustrates the geospatial distribution of the collected samples, highlighting country-level coverage and total PV area. We further conducted a statistical analysis of PV instance scales across varying resolutions, categorizing them into tiny (0–100 m²), small (100–400 m²), medium (400–900 m²), and large (> 900 m²). Tiny PV panels represent the most frequent category in high-resolution imagery (< 0.3 m), accounting for 75,578 samples. Small- and medium-scale systems are less prevalent, comprising 28,153 and 23,571 instances, respectively, and are primarily identified in imagery ranging from 0.1 m to 10 m. The most abundant category is large-scale PV installations, consisting of 149,519 samples, the vast majority of which are captured by Sentinel-2 imagery at 10 m and 20 m resolutions.

3.2. PV-specific binary prompt

To overcome the scarcity of dense captions, we adopt a practical trade-off by employing a binary prompt strategy that explicitly distinguishes PV-present from PV-absent imagery. While detailed captions (e.g., “A 0.1m-resolution satellite image showing dark PV panels on an urban rooftop”) theoretically maximize semantic granularity, this pattern establishes an efficient and robust baseline for developing a PV-specific VLM. Crucially, although the semantic scope is constrained to presence or absence, the underlying contrastive alignment forces the model to ground these concepts spatially. Consequently, the model learns to associate the text “PV panels” with specific visual

textures, contexts, and geometries. It retains the core advantage of the VLM paradigm – robust text-to-patch spatial alignment – enabling meaningful zero-shot capabilities for dense prediction tasks.

As illustrated in Fig. 4, we constructed a structured vocabulary library categorized into RS imagery descriptors, Presence/Absence indicators, and PV-related terms. During training, we dynamically generate unique captions by randomly combining terms from these categories. Specifically, positive and negative samples are distinguished solely by their respective indicators (e.g., “a satellite image containing solar panels” vs. “an aerial imagery without PV infrastructure”). This randomized generation strategy introduces syntactic diversity, effectively preventing the model from overfitting to static sentence structures and encouraging robust vision-language alignment for PV semantics.

3.3. PANEL pre-training dataset construction

Building upon the diverse worldwide imagery and the binary prompt strategy, we construct the final pre-training dataset for PANEL. This process involves pairing the raw RS imagery with dynamically text descriptions to form aligned vision-language pairs. To ensure the model learns robust representations across varying scales and contexts, we apply a standardized preprocessing pipeline to transform the heterogeneous source images into a uniform format suitable for CLIP training.

To prepare the image data, we adopt a deliberate cropping strategy designed to reduce redundancy while preserving the integrity of small-scale distributed PV panels. Specifically, images larger than 480 pixels are cropped using a 256 × 256 sliding window with a 224-pixel stride (12.5% overlap). This overlap minimizes the risk of fragmenting PV objects or losing critical contextual information at the boundaries. Conversely, images smaller than 480 pixels are directly resized to 256 × 256. Following this processing, each image crop can be categorized as PV-present (positive) or PV-absent (negative) based on ground truth annotations and is then paired with a matching binary prompt randomly sampled from our structured vocabulary library.

In total, this pipeline yields a substantial number of 1,040,693 image-text pairs, comprising 593,438 PV-present and 447,255 PV-absent samples. To facilitate a rigorous and fair evaluation of zero-shot capabilities – specifically for image classification and semantic localization tasks – we randomly reserve 5% of these processed samples as a

Table 2
Current available PV datasets in RS.

Dataset	Source	Region	Spatial resolution	With imagery	Image size	Count	Interpretation task	Used in this study
BDAPPV (Kasmi et al., 2023)	Google Earth; IGN	France	0.1 m; 0.2 m	✓	400	28,807; 17,325	Segmentation	✓
Heilbronn in Germany (Li et al., 2021)	Google Earth	Heilbronn	0.15 m	✓	13,398	40	Segmentation	✓
Southern Germany (Clark and Pacifici, 2023)	WorldView-3; Maxar Technologies	Nuremberg; Regensburg; Munich	0.31 m; 0.155 m;	✓	416; 832;	2542; 2542;	Object detection	✓
PV 01/03/08 (Jiang et al., 2021)	Gaofen-2; Beijing-2; UAV	Jiangsu, China	0.1 m; 0.3 m; 0.8 m	✓	256; 1024; 1024	645; 2308; 763	Segmentation	✓
California in USA (Bradbury et al., 2016)	USGS orthoimagery data	Fresno; Modesto; Stockton; Oxnard	0.3 m; 0.3 m; 0.3 m; 1 m	✓	~ 5,000	412; 20; 94; 75	Segmentation	✓
AIR-PV (Yan et al., 2023)	Gaofen-2	Western China	0.8 m	✓	512	5000	Segmentation	✓
China PV-2020 (X. Zhang et al., 2022)	Sentinel-2	China	30 m	✗	—	—	Segmentation	✗
China PV-2022 (Chen et al., 2024)	Sentinel-2	China	10 m	✗	~400 to ~1000	29,989	Segmentation	✓
Solar farms in India-2021 (Ortiz et al., 2022)	OSM; Sentinel-2	India	10 m	✗	—	—	Segmentation	✗
China PV-2018 (Kruitwagen et al., 2021)	Sentinel-2; SPOT6/7	Worldwide	10 m; 1.5 m	✗	—	—	Segmentation	✗
Global wind/solar farms-2020 (Dunnnett et al., 2020)	OSM	Worldwide	—	✗	—	—	—	✗
Global PV-2022 (A. Li et al., 2025)	Google Earth; Sentinel-2	Worldwide	0.6 m; 20 m	Partly; ✗	~1690; ~400 to ~1350	179; 95,110	Segmentation	✓

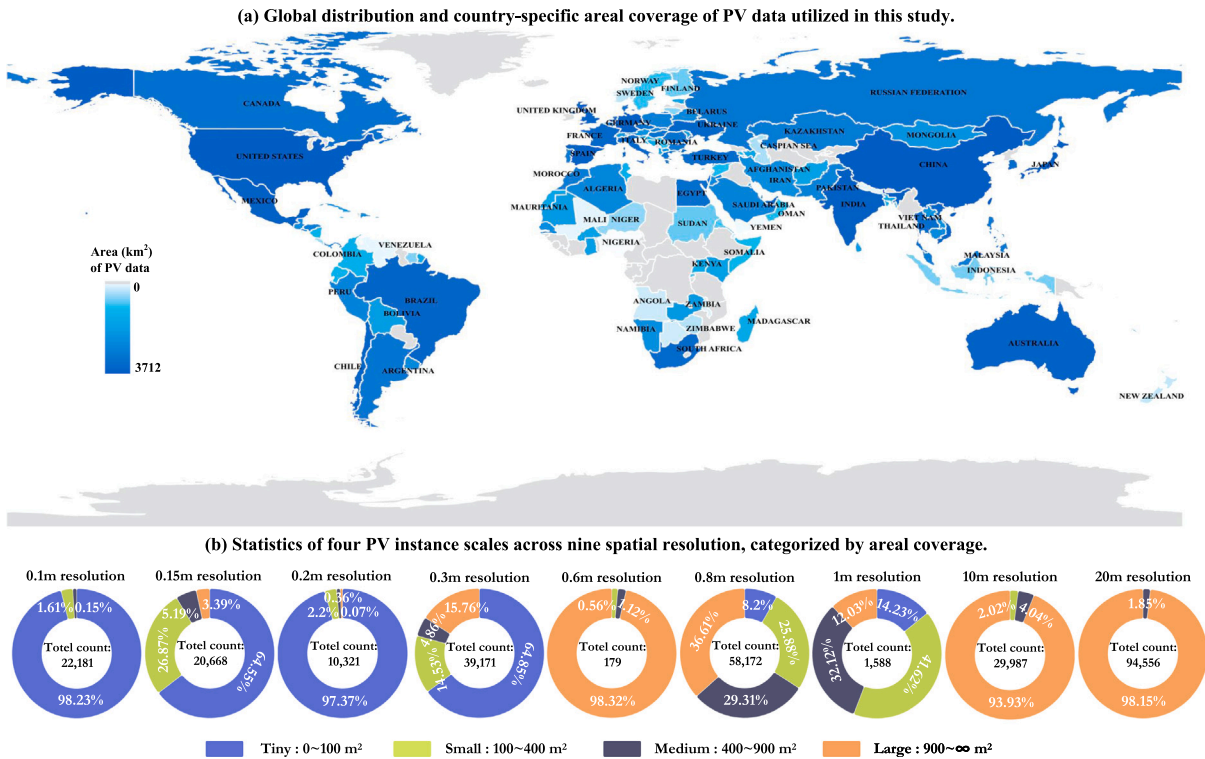


Fig. 3. Global distribution and relevant statistical analysis of PV data utilized in this study.

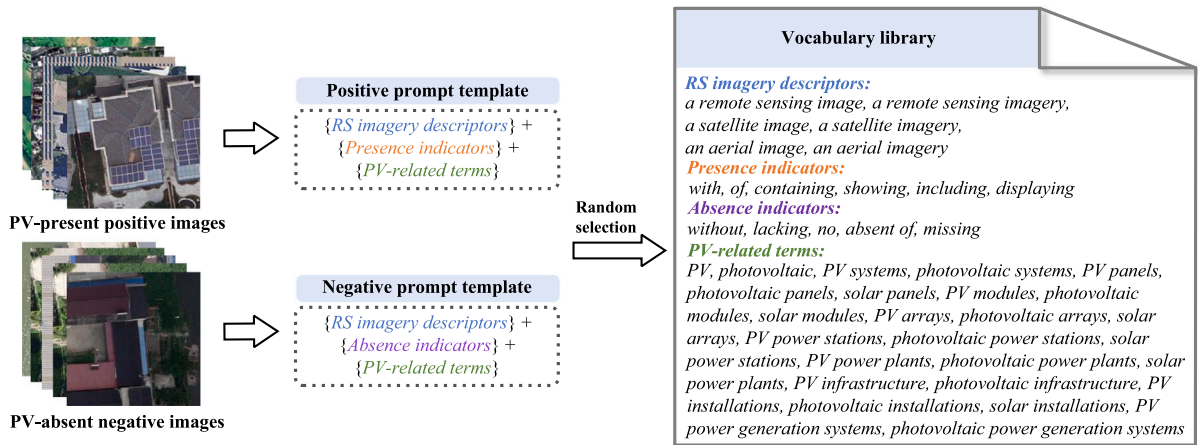


Fig. 4. Prompts for PV-present positive and PV-absent negative images through corresponding templates to randomly select terms from the vocabulary library.

Table 3

Specifications of the seven PV-specific benchmarks established in this study.

Benchmark	Task	Resolution (m)	Size (px)	Source composition	Count
PV-Cls	Classification	0.1, 0.15, 0.2, 0.3, 0.6, 0.8, 1.0, 10 and 20	256 to 480	Random 5% of each dataset	29,671 positives and 22,362 negatives
PV-SL	Semantic localization	0.1, 0.15, 0.2, 0.3, 0.6, 0.8, 1.0, 10 and 20	256 to 480	Positive samples of PV-Cls and exclude Southern Germany	28,511 positives
PV-SegA	Segmentation	0.1, 0.15, 0.2 and 0.3	400	BDAPPV, Heilbronn, PV 01/03, Fresno, Modesto, and Stockton	48,962 positives
PV-SegB	Segmentation	0.6, 0.8 and 1.0	512	Global PV-2022 (0.6 m), PV 08, Oxnard, and AIR-PV	6034 positives
PV-SegC	Segmentation	10 and 20	256	China PV-2022 and Global PV-2022 (20 m)	507,456 positives
PV-Det	Object detection	0.155	832	Southern Germany	2542 positives
PV-Gen	Image generation	0.1, 0.15, 0.2, and 0.3	400	PV-SegA	48,962 positives

dedicated test set. Consequently, the final PANEL pre-training dataset consists of 988,658 pairs.

3.4. Downstream task benchmarks

To rigorously evaluate PANEL against generalist VLMs, we established seven distinct benchmarks covering five core PV-relevant interpretation tasks, including classification, semantic localization, segmentation, object detection, and image generation. Unlike the pre-training dataset, these benchmarks were constructed with different pre-processing protocols to ensure fair comparison across varied tasks. The detailed specifications for each benchmark are presented in Table 3.

PV-Cls: For image classification, this benchmark is directly derived from the 5% holdout set reserved during the pre-training dataset construction phase. It is specifically designed to evaluate the zero-shot generalization capabilities of different CLIP models. While the original sample dimensions range from 256 to 480 pixels, during testing, all images are resized and normalized to match the standard input format specifications required by each respective CLIP model.

PV-SL: The semantic localization benchmark is constructed by filtering the PV-Cls dataset to exclude Southern Germany data (Clark and Pacifici, 2023) and retain only positive samples. In contrast to the classification task, we do not resize these images during evaluation. Instead, we preserve their original dimensions (ranging from 256 to 480 pixels) to assess the model's ability to localize targets within their native spatial contexts and varying scales.

PV-SegA/B/C: For segmentation tasks, we constructed three distinct datasets stratified by specific resolution bands. To maintain the

inherent characteristics of each source, crop sizes were selected to align with major existing datasets:

- **PV-SegA** (0.1–0.3 m): Uses a crop size of 400 pixels, aligning with the standard of the BDAPPV dataset (Kasmi et al., 2023).
- **PV-SegB** (0.6–1.0 m): Uses a crop size of 512 pixels, referencing the AIR-PV dataset (Yan et al., 2023).
- **PV-SegC** (10–20 m): Uses a crop size of 256 pixels, referencing the China PV-2022 dataset (Chen et al., 2024).

All large-scale source imagery is cropped using these specific window sizes with a stride ensuring 12.5% overlap. Notably, for these segmentation benchmarks, we focus exclusively on positive samples to better assess model performance on PV-relevant regions, as negative samples (pure background) typically yield near-perfect accuracy for all methods.

PV-Det: The object detection benchmark is composed exclusively of the Southern Germany dataset (0.155 m resolution, 832-pixel size). We deliberately avoid converting other segmentation datasets into detection benchmarks due to the inherent limitations and potential label noise associated with deriving bounding boxes from instance masks. Similar to segmentation, this benchmark only evaluates positive samples.

PV-Gen: For the image generation benchmark, our study specifically evaluates the capability of CLIP models to condition generative processes within urban scenarios dominated by rooftop distributed PV panels. Consequently, we constructed the dataset following the same protocol as PV-SegA, targeting spatial resolutions ranging from 0.1 m to 0.3 m where distributed PV installations are most prevalent, resulting

in a total of 48,962 PV-positive samples. To ensure consistency during generation, we employ a standardized set of prompts tailored to these urban PV scenes, for example: “a high-resolution remote sensing image of urban rooftops with clearly visible photovoltaic panels under natural daylight conditions”. In parallel, we apply a unified negative prompt—“blurry, low quality, low resolution, bad anatomy, bad proportions, deformed, disfigured, distorted”—to suppress undesired artifacts and enforce higher-fidelity outputs.

4. Methodology

4.1. Pre-training pipeline for panel

We selected the ViT-B/16 version of CLIP from the field of computer vision as our base model, where the image encoder employs a ViT with a patch size of 16 and a base parameter configuration. This version has the highest accuracy among efficient models while maintaining a significantly lower parameter count than larger variants (e.g., ViT-L/14) (Y. Li et al., 2025). Through cross-modal supervision on our constructed large-scale image-text pairs, PANEL learns to align the visual features of PV panels with their corresponding binary prompts, effectively grouping PV-related semantics while pushing apart representations of PV-absent samples.

Formally, we define our worldwide PV vision-language dataset as $D = \{(x_i^I, x_i^T)\}_{i=1}^N$, where N denotes the total number of training samples, and x_i^I and x_i^T represent the i -th image and text inputs, respectively. Our optimization objective is to learn an image encoder f_p^I and a text encoder f_p^T that can extract their aligned latent representation, i.e., $f_p^I(x_i^I) = p_i^I \in \mathbb{R}^{d_p \times 1}$ and $f_p^T(x_i^T) = p_i^T \in \mathbb{R}^{d_p \times 1}$. The extracted features are projected into a shared embedding space with dimension $d_p = 512$. During pre-training, PANEL formulates an instance discrimination task within each batch and optimizes the following bi-directional InfoNCE objective:

$$\mathcal{L}_{\text{InfoNCE}} = - \left(\underbrace{\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(p_i^I \cdot p_i^T / \tau)}{\sum_{j=1}^N \exp(p_i^I \cdot p_j^T / \tau)}}_{\text{image to text}} + \underbrace{\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(p_i^T \cdot p_i^I / \tau)}{\sum_{j=1}^N \exp(p_i^T \cdot p_j^I / \tau)}}_{\text{text to image}} \right) / 2 \quad (1)$$

where τ is a learnable temperature parameter initialized to 0.07.

To better adapt the model to small-scale distributed PV panels, we re-interpolated the original CLIP positional embeddings using bicubic interpolation. With a patch size of 16, this modification increases the output feature map dimensions from 14×14 to 16×16 , enabling the model to capture more fine-grained PV-relevant features.

Furthermore, to address the inherent variability of RS imagery, we employ a comprehensive augmentation pipeline designed to significantly enhance the model’s resilience and generalization capabilities in challenging scenarios. Specifically, this pipeline incorporates a diverse suite of techniques tailored to PV imagery, including geometric transformations (flipping, rotation, scaling, and translation), photometric adjustments (brightness, contrast, saturation, hue, and haze), and the simulation of real-world artifacts (cloud occlusion, motion blur, Gaussian blur and noise, compression artifacts, and salt-and-pepper noise).

4.2. PANEL surgery for PV-relevant priors extraction

Standard CLIP models often exhibit a bias towards background regions rather than foreground objects, frequently resulting in noisy activations at irrelevant positions during visualization. Y. Li et al. (2025) attribute these inconsistencies to the standard q - k - v self-attention mechanism, which can induce semantically misaligned token interactions

and amplify redundant category-agnostic features. To mitigate this, they introduced CLIP Surgery, a method that incorporates a parallel branch based on a v - v - v self-attention mechanism to extract patch-level image features. This auxiliary branch shares parameters directly with the original q - k - v path and serves as an explainability tool capable of positively activating regions corresponding to specific class prompts without requiring any fine-tuning. This establishes a simple yet effective paradigm for zero-shot semantic localization.

Building upon this architecture, we propose PANEL Surgery, which is explicitly designed to extract three distinct types of PV-relevant priors – the global similarity score, the spatial similarity map, and the original patch-level q - k - v image features – to guide downstream zero- and few-shot learning, as shown in Fig. 5(a). Specifically, we augment the pre-trained PANEL image encoder with a parallel transformer branch that employs a consistent v - v - v attention mechanism. This parallel branch shares all parameters (MLP and LayerNorm) with the original branch, except for the attention projection weights which are re-routed to use value (v) for query and key projections. Given an input image $x_i^I \in \mathbb{R}^{3 \times H \times W}$, the original and parallel branches respectively produce flattened spatial feature maps $p_i^{s_o}, p_i^{s_n} \in \mathbb{R}^{(H_{16}W_{16}) \times d_p}$, where H_{16} and W_{16} are the image height and width divided by the patch size, and d_p is the feature embedding dimension. The class token features $p_i^I \in \mathbb{R}^{d_p \times 1}$, carrying global visual semantics, are simultaneously extracted for image-level understanding. Notably, PANEL Surgery supports flexible image resolutions via bicubic interpolation of positional embeddings, avoiding the default resizing to training size, and preserving critical PV texture details.

The text encoder processes both the full prompt x_i^T and a redundant prompt $x_i^{T_r}$ to derive a positive text feature $p_i^{T_p} \in \mathbb{R}^{d_p \times 1}$ and a redundant text feature $p_i^{T_r} \in \mathbb{R}^{d_p \times 1}$, respectively. The redundant prompt is constructed by strictly removing PV-specific class tokens (e.g., “photovoltaic”, “solar panel”, “PV array”) from the template while retaining context words (e.g., “a satellite image of...”). As $p_i^{T_r}$ captures background semantics that often introduce noisy activations, we subtract it to obtain a purified and PV-focused text representation, i.e., $p_i^{T_v} = p_i^{T_p} - p_i^{T_r}$. We then compute both a global similarity score and a spatial similarity map via cosine similarity:

$$sm_i^g = \frac{p_i^I \cdot (p_i^{T_v})^T}{\|p_i^I\| \times \|p_i^{T_v}\|} \quad (2)$$

$$sm_i^s = \frac{p_i^{s_n} \cdot (p_i^{T_v})^T}{\|p_i^{s_n}\| \times \|p_i^{T_v}\|}$$

The global similarity score $sm_i^g \in \mathbb{R}^{1 \times 1}$ measures the similarity between the input image and the text, and is typically used in image-level tasks such as image retrieval and classification. A larger value of sm_i^g indicates a higher likelihood that the image is a PV-present positive sample. The spatial similarity map $sm_i^s \in \mathbb{R}^{H_{16} \times W_{16} \times 1}$ measures the similarity between patch-level features and text features along the spatial dimension, where a larger value at a given position suggests a higher probability of PV-relevant semantics. Furthermore, the image features $p_i^{s_o}$ produced by the original branch do not involve a cosine similarity computation with the text features, because the parallel branch has greater vision-language alignments. However, in terms of image feature extraction, $p_i^{s_n}$ from the parallel branch discards the contributions of k and q present in the q - k - v self-attention of the original branch. We therefore regard $p_i^{s_o}$ as capable of capturing generalizable features that are valuable for few-shot adaptations.

4.3. PANEL for zero-shot inference

Based on PANEL Surgery, Fig. 5(b) illustrates the architecture for implementing various zero-shot inference tasks. All parameters of PANEL Surgery and other associated networks are frozen, with no fine-tuning involved. Inference on downstream tasks is performed directly through further computations of output features from PANEL Surgery.

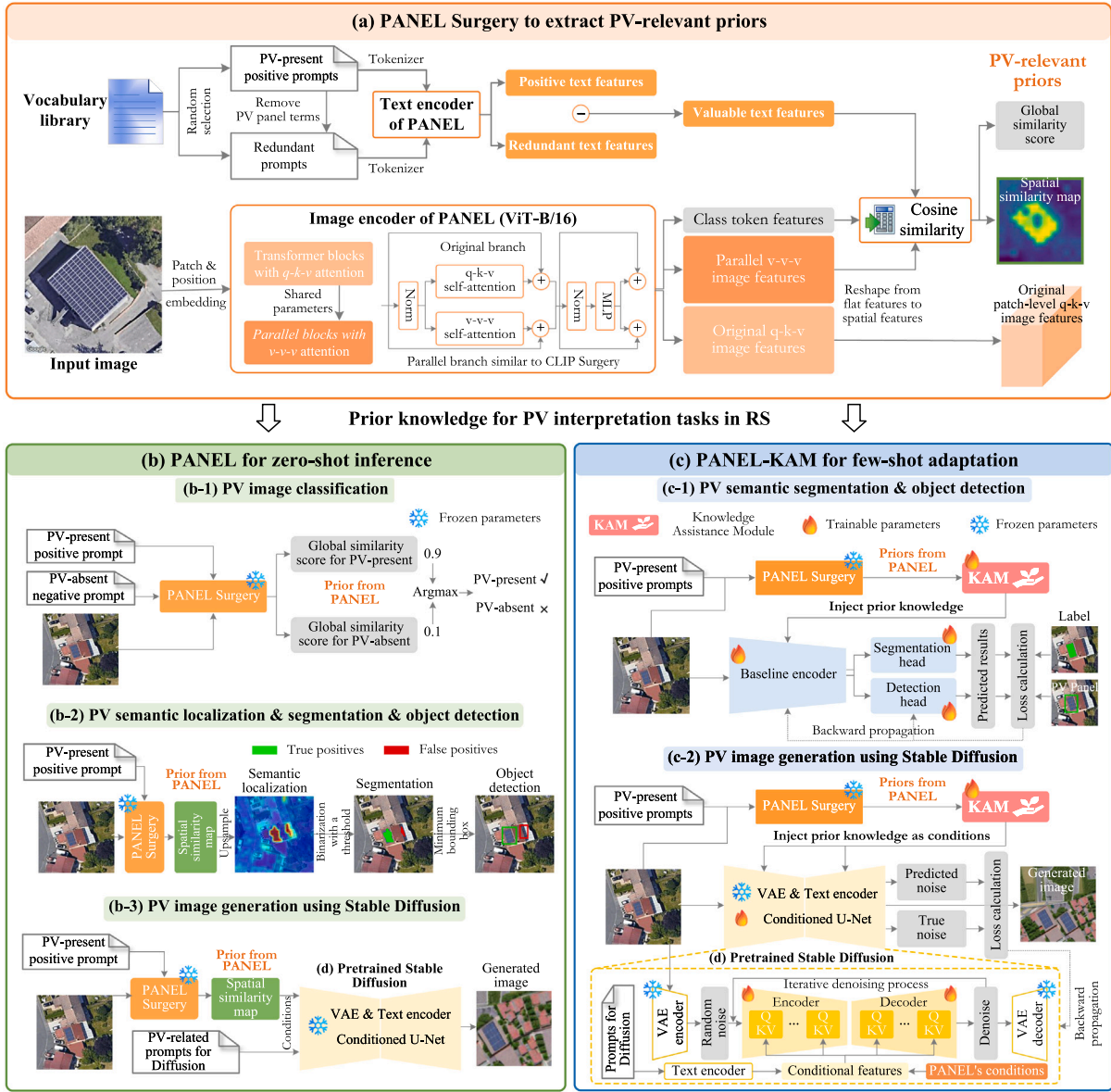


Fig. 5. Overall framework of PANEL for diverse PV-related downstream tasks. (a) PANEL Surgery extracts PV-relevant priors (global similarity score, spatial similarity map, and patch-level features) via a parallel $v-v-v$ attention branch. (b) For zero-shot inference, these priors directly support training-free image classification, semantic localization, segmentation, detection, and generation. (c) For few-shot adaptation, the proposed Knowledge Assistance Module (KAM) injects these priors to enhance trainable baselines across various tasks.

As for PV image classification, we determine whether an input image is PV-present or PV-absent by comparing its responses to positive and negative prompts. Specifically, both prompts are tokenized and encoded by PANEL's text encoder to generate positive text features $p_i^{T_p}$ and negative text features $p_i^{T_n}$. These are then used to compute the cosine similarity with the global class token feature p_i^l from the image encoder. The influence of redundant prompts is ignored in this task, as the key information for classification lies in the presence or absence indicators within the prompts. By comparing the global similarity scores sm_i^g under positive and negative prompts, PANEL can classify whether the image contains PV panels.

For PV semantic localization, segmentation, and object detection, all tasks are fundamentally based on the spatial similarity map sm_i^s computed by PANEL Surgery. After confirming that an image is PV-present via sm_i^g , PANEL Surgery calculates the cosine similarity between the valuable text features $p_i^{T_v}$ and the spatial features $p_i^{s_n}$ generated by the parallel branch with $v-v-v$ attention. This yields the PV-related

activation map $sm_i^s \in \mathbb{R}^{H_{16} \times W_{16} \times 1}$, which is then upsampled via bicubic interpolation to the original image size for final PV semantic localization. Subsequently, binarizing the upsampled sm_i^s with a threshold and extracting the minimum bounding rectangle enables PV segmentation and object detection.

For PV image generation, directly synthesizing panels is difficult due to complex geometry, specular reflections, and fine textures. Therefore, we adopt an inpainting approach to enrich backgrounds while preserving PV-support regions. Specifically, a spatial similarity map sm_i^s is upsampled to yield a binary mask $M_i \in \mathbb{R}^{H \times W}$ that isolates editable background from PV-support areas. The input image x_i^l is encoded by the VAE, $z_i = \mathcal{E}(x_i^l) \in \mathbb{R}^{H_8 \times W_8 \times c}$. The mask is downsampled and broadcast to latent resolution, $M_i' \in \mathbb{R}^{H_8 \times W_8 \times c}$, and we define the masked latent $z_i' = z_i \odot (1 - M_i')$, fixing PV regions and exposing only background for editing. Guided by text embeddings $\tau_\theta(\cdot)$ via cross-attention, a frozen Stable Diffusion Inpainting U-Net is used to iteratively denoise z_i' . Following T denoising steps, we obtain the reconstructed latent $\hat{z}_i$ followed by the VAE decoder to obtain

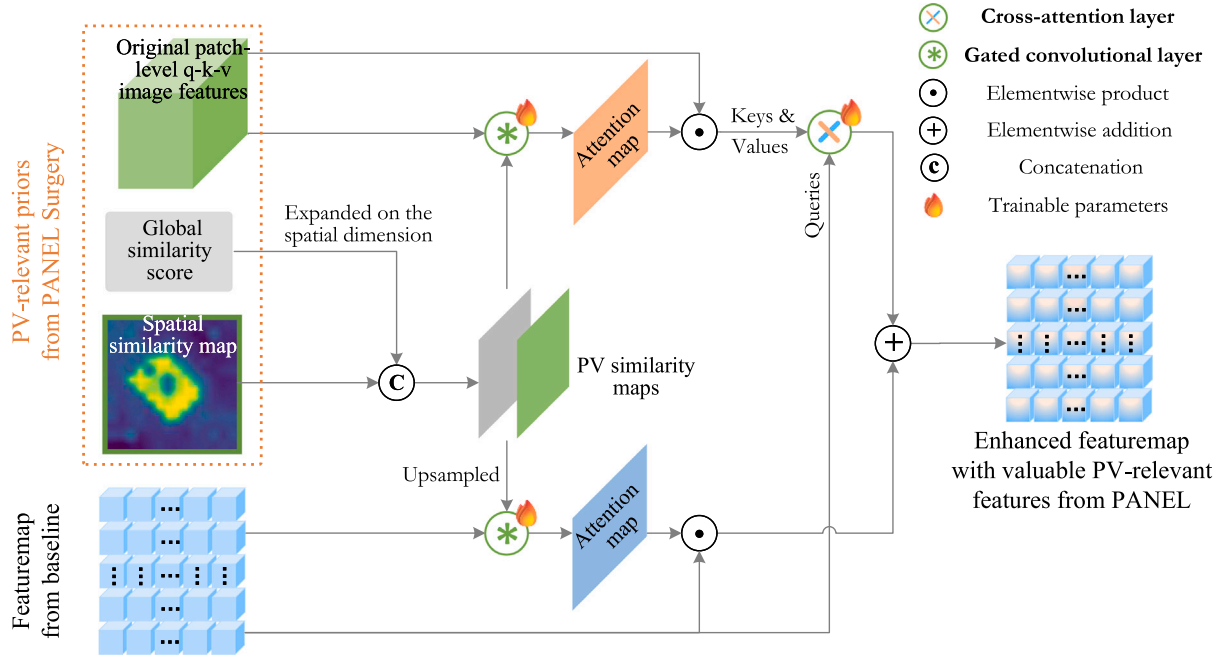


Fig. 6. Architecture of our proposed Knowledge Assistance Module (KAM) with gated convolution and cross-attention fusion to enhance baselines' semantics via imparting PV-relevant priors from PANEL.

$\hat{x}_i = D(\hat{z}_i) \in \mathbb{R}^{H \times W \times 3}$, producing diverse, realistic backgrounds with PV-support regions strictly preserved.

4.4. PANEL-KAM for few-shot learning

Although PANEL can support zero-shot inference, its frozen representations lack the task-specific decoding mechanisms required for dense prediction tasks, such as segmentation, detection, and image generation. Consequently, adapting dedicated baselines to these downstream tasks necessitates task-specific fine-tuning, especially when confronting domain shifts across unseen regions, diverse sensors, or varying spatial resolutions. However, in data-scarce scenarios, these baselines often suffer from severe overfitting and fail to learn discriminative PV features. Therefore, the critical limitation post-pre-training is how to effectively transfer PANEL's robust PV semantics into downstream architectures under limited supervision. To bridge this gap, we propose the KAM module. KAM acts as a plug-and-play bridge that injects the PV-relevant priors extracted by PANEL Surgery into trainable baselines, guiding their convergence and significantly enhancing few-shot performance.

Specifically, as illustrated in Fig. 6, we utilize the three PV-relevant features extracted from PANEL Surgery and adaptively integrate them into any featuremap of the baseline model via our proposed KAM module. First, the global similarity score sm_i^g is expanded along the spatial dimension and concatenated with the spatial similarity map sm_i^s to form $sm_i^{all} \in \mathbb{R}^{H_{16} \times W_{16} \times 2}$. Based on sm_i^{all} , two gated convolutional layers (⊗) are applied to generate attention maps α_p and α_b for the original spatial q-k-v features $p_i^{s_o}$ and the baseline featuremap b_i , respectively. These attention maps enable adaptive information selection to enhance PV-relevant representations in both $p_i^{s_o}$ and b_i , thus aiding the baseline in learning PV-relevant semantics under few-shot conditions. The computation of α_p and α_b is as follows:

$$\alpha_p = p_i^{s_o} \otimes sm_i^{all} = \sigma(Convl_{1 \times 1}(p_i^{s_o} || sm_i^{all})) \quad (3)$$

$$\alpha_b = b_i \otimes sm_i^{all} = \sigma(Convl_{1 \times 1}(b_i || Upsample(sm_i^{all}))) \quad (4)$$

Here, σ denotes the sigmoid activation function, $Convl_{1 \times 1}$ acts as a projection layer mapping channels to 1, and $||$ indicates feature concatenation. Since b_i can be from any layer of the baseline and thus

varies in size, sm_i^{all} is upsampled via bicubic interpolation to match the spatial resolution of b_i . The attention map α_p allows $p_i^{s_o}$ to suppress irrelevant activations via $p_i^{s_o} * \alpha_p$.

Next, we use a cross-attention fusion layer to further adaptively extract valuable features from the gated $p_i^{s_o}$ and impart them to the baseline feature map b_i . The computation is as follows:

$$b_i^{ca} = \frac{QK^T}{\sqrt{d_p}} V \quad (5)$$

$$Q = Conv1D(Reshape(b_i))$$

$$K = Conv1D(Reshape(p_i^{s_o} * \alpha_p)) \quad (6)$$

$$V = Conv1D(Reshape(p_i^{s_o} * \alpha_p))$$

Here, b_i^{ca} is the output of this layer, d_k is the scaling factor, and $Conv1D$ denotes a 1×1 depth-wise projection. The final output feature map after the KAM module is obtained via a residual connection:

$$b_i^{KAM} = b_i + b_i * \alpha_b + b_i^{ca} \quad (7)$$

For few-shot PV segmentation and object detection tasks, we recommend inserting the KAM module into multiple encoder layers of the baseline model. Since the semantic representations of b_i differ across layers—with low-level features capturing fine textures and high-level features encoding rich semantics—the KAM module enables each layer to adaptively incorporate different PV-relevant priors from PANEL. During training, all KAM modules and baseline parameters are trainable, while the parameters of PANEL remain frozen.

For few-shot image generation, we build upon the Stable Diffusion inpainting framework by fine-tuning its denoising U-Net. Specifically, the enhanced representation b_i^{KAM} is utilized as spatial priors to ensure panel preservation and increase background diversity. These representations are injected into the U-Net via cross-attention to minimize boundary distortions while enhancing stability and fidelity during few-shot training. To offer the model clearer region guidance, we also expand the input channels of the U-Net by concatenating the original inputs with the binary mask M'_i . This allows for the explicit separation of editable backgrounds and preserved PV areas, which stabilizes boundary synthesis while keeping the pre-trained backbone intact. Following the standard linear variance schedule with $T = 1000$ steps,

the denoising U-Net ϵ_θ is conditioned jointly on text prompts and the KAM-enhanced features. The training objective is defined as a masked noise-prediction loss:

$$\mathcal{L} = \mathbb{E}_{\mathcal{E}(x_t^I), \epsilon \sim \mathcal{N}(0,1), t} \left[\left\| \epsilon - \epsilon_\theta \left(z'_{i,t}, t; \tau_\theta(y), b_i^{KAM}, M'_i \right) \right\|_2^2 \right], \quad (8)$$

where $\mathcal{E}$ denotes the pre-trained VAE encoder, $z'_{i,t}$ is the noisy latent at diffusion step t , $\tau_\theta(y)$ represents the text encoder output of prompt y , ϵ is the ground truth noise, and M'_i supplies explicit region guidance. During inference, Classifier-Free Guidance (CFG) with scale w is employed to balance synthesis fidelity and prompt alignment.

5. Experimental results and analysis

5.1. Experimental settings

5.1.1. Comparative methods

To comprehensively evaluate the efficacy of PANEL, we benchmark it against five representative CLIP models: CLIP (Radford et al., 2021) and four variants in RS—RemoteCLIP (Liu et al., 2024), Sky-CLIP (Z. Wang et al., 2024), GeoRSCLIP (Zhang et al., 2024), and Git-RSCLIP (Liu et al., 2025). For each baseline, we evaluate all publicly available checkpoints to assess their zero-shot performance across five downstream PV-relevant tasks: image classification, semantic localization, segmentation, object detection, and image generation.

For few-shot learning scenarios, we integrate our proposed PANEL and KAM module into diverse baseline architectures to test generalization. For PV segmentation, we select four baselines representing different paradigms from MMSegmentation: MaskFormer (Cheng et al., 2021) (mask classification), SegFormer (Xie et al., 2021) (transformer-based), OCRNet (Yuan et al., 2021) (object-contextual representations), and DeepLabV3+ (Chen et al., 2018) (encoder-decoder with atrous convolution). For PV detection, we employ three widely used detectors from MMDetection: DINO (H. Zhang et al., 2022) (DETR-based Carion et al., 2020), YOLOX (Ge et al., 2021) (anchor-free one-stage), and Faster R-CNN (Ren et al., 2016) (two-stage). To ensure fair comparison and focus on the efficiency of few-shot adaptation, we utilize the most lightweight backbone versions for all baselines. For image generation, we utilize Stable Diffusion 3 to evaluate the quality of samples conditioned on the priors extracted from the aforementioned CLIP models.

5.1.2. Implementation details

PANEL Pre-training: We initialize PANEL with the pre-trained weights of CLIP (ViT-B/16). The model is trained end-to-end using the AdamW optimizer with a learning rate of 1×10^{-6} and a weight decay of 0.01. Training is conducted on three NVIDIA A100 (80 GB) GPUs with a batch size of 256 per GPU. We train for a maximum of 100 epochs, employing an early stopping mechanism with a patience of 5 epochs based on validation loss. The total training time was approximately 2 days. As detailed in Section 4.1, we apply a comprehensive suite of augmentations during training to enhance robustness against RS-specific variations.

Zero-shot Evaluation: Inference is performed by computing the cosine similarity between the image embeddings and text embeddings extracted by each model. To ensure a fair comparison, the same set of positive and negative prompts (enumerated from our vocabulary library) is used across all comparative methods.

Few-shot Adaptation: We follow the standard training protocols provided in the MMSegmentation and MMDetection frameworks. In these experiments, the parameters of PANEL are frozen, and only the baseline networks and the inserted KAM modules are fine-tuned on the few-shot subsets. The hidden layer dimensions of the KAM module are dynamically adjusted to match the channel dimensions of the corresponding baseline feature maps. Additionally, during the training phase, we use a dynamic prompting strategy where a single

text prompt is randomly generated from the vocabulary library for each iteration. For inference, however, we employ an ensemble strategy: we enumerate all possible positive and negative prompt combinations from the library and average the resulting similarity maps. This ensemble approach mitigates the potential bias introduced by any single prompt, ensuring more stable and robust guidance.

5.2. Evaluation metrics

For classification, segmentation, and object detection tasks, we report standard performance metrics including accuracy, Intersection over Union (IoU), precision, recall, F1-score, and AP@0.5/ 0.5:0.95 (average precision at IoU threshold 0.5 and averaged over multiple IoU thresholds from 0.5 to 0.95 in steps of 0.05).

For semantic localization, we utilize the mean Score Contrast (mSC) metric (Y. Li et al., 2025). mSC measures the difference in activation intensity between the foreground and background regions. It ranges from -100% to 100% and is defined as:

$$\text{mSC} = \frac{1}{N} \sum_i [m_p(sm_i^s \cdot G) - m_p(sm_i^s \cdot \neg G)] \quad (9)$$

where sm_i^s is the similarity map for the i th sample, G is the corresponding binary foreground ground-truth mask, $\neg G$ denotes the background ground-truth mask, and N is the number of test images. The operator m_p computes the mean values over pixels (spatial average).

To evaluate the performance of image generation, we use the CLIP score (Hessel et al., 2022), which measures the cosine similarity between the image and its corresponding text prompt using the original CLIP (ViT-B/16). Since the value of the CLIP score is relative, we report a normalized metric, Δ_{CLIP} . This metric represents the mean absolute difference between the generated scores and the real mean, normalized by the absolute value of the real mean to provide a scale-invariant comparison. It is defined as:

$$\Delta_{CLIP} = \left(\frac{1}{M \cdot |\bar{s}_{real}|} \sum_{j=1}^M |s_j^{gen} - \bar{s}_{real}| \right) \times 100\% \quad (10)$$

where s_j^{gen} is the CLIP score of the j th generated image, $\bar{s}_{real}$ is the mean CLIP score of all real images, and M is the number of generated images. A smaller Δ_{CLIP} indicates that the generated images are more similar to real images in terms of vision-language alignment.

5.3. Assessments on zero-shot downstream tasks

5.3.1. Image classification

Table 4 details the zero-shot classification accuracy of PANEL and representative RS CLIP models across nine distinct spatial resolution bands ranging from 0.1 m to 20 m. PANEL demonstrates consistently superior robustness across the entire spectrum, maintaining high accuracy (89.73%–97.96%) in both high-resolution imagery dominated by distributed rooftop PV panels and low-resolution satellite data containing large-scale solar farms. In contrast, generalist CLIP models exhibit significant volatility.

A critical gap is observed in the low-resolution 10 m and 20 m subsets, where the test data is heavily imbalanced towards positive samples (e.g., 6497 positive vs. 55 negative in the 10 m subset; 17,691 positive vs. 1171 negative in the 20 m subset). In this context, the extremely low accuracy scores reported for models such as GeoRSCLIP (2.99%) and RemoteCLIP ViT-B/32 (3.48%) do not merely reflect poor performance but indicate a degenerate prediction pattern where the models fail to recognize the spectral signatures of solar farms. While some generalist CLIP models perform adequately on high-resolution imagery, their complete failure on these coarser resolutions further confirms that they lack exposure to PV-specific samples and relevant semantic descriptions during pre-training. Conversely, PANEL achieves 97.50% and 92.41% accuracy on these subsets, proving its capability to overcome this “single-class collapse” and effectively identify PV targets across diverse sensors and resolutions.

Table 4

Zero-shot PV image classification accuracy (%) of PANEL and other representative CLIP models across varying spatial resolutions (0.1 m to 20 m). The highest values are highlighted in bold for each resolution band.

Method	Version	0.1 m	0.15 m	0.2 m	0.3 m	0.6 m	0.8 m	1 m	10 m	20 m
CLIP	ViT-B/16	61.84	84.95	67.29	76.20	79.50	65.92	75.81	61.55	40.28
	ViT-L/14	62.80	74.48	66.94	76.43	77.01	64.72	85.71	51.05	33.25
	ResNet-50	67.19	85.02	58.18	39.58	63.71	50.54	39.40	86.71	81.33
RemoteCLIP	ViT-B/32	54.08	84.62	55.02	82.55	70.08	75.15	94.47	3.48	6.57
	ViT-L/14	52.51	84.42	55.37	83.20	68.42	71.64	94.70	1.37	6.69
	ViT-B/32	65.68	78.22	53.97	56.39	79.50	65.65	73.50	87.90	62.95
SkyCLIP	ViT-L/14	58.20	88.18	58.64	89.69	82.83	88.33	94.24	10.49	13.91
	ViT-B/32	62.18	86.08	59.35	77.58	67.87	73.18	94.47	2.99	6.71
	ViT-L/14	53.12	85.17	57.01	85.78	68.98	73.65	94.93	6.33	8.81
GeoRSCLIP	ViT-H/14	52.99	84.47	55.26	85.03	68.98	72.84	95.16	5.88	9.81
	ViT-B/16	66.85	52.45	64.84	89.89	79.50	85.63	84.33	8.85	12.28
	ViT-L/16	56.90	77.32	60.40	88.50	72.85	81.38	87.56	8.70	11.25
PANEL (ours)	ViT-B/16	95.16	97.96	94.27	93.90	89.73	95.78	96.93	97.50	92.41

Table 5

Zero-shot PV semantic localization performance of PANEL and representative RS CLIP models on the PV-SL benchmark, evaluated by the mean Score Contrast (mSC) metric. The highest values are highlighted in bold at each subset.

Method	Version	BDAPPV	Heilbronn	PV 01/03/08	California	AIR-PV	China PV-2022	Global PV-2022	Overall PV-SL
CLIP	ViT-B/16	−29.62	−19.49	−13.72	−17.28	−11.67	0.04	−2.11	−3.56
	ViT-L/14	−27.84	−18.29	−12.66	−20.24	−11.23	1.75	0.57	−1.35
CLIP Surgery	ViT-B/16	37.89	34.42	34.86	32.95	17.05	3.82	0.78	5.06
	ViT-L/14	42.12	29.24	23.54	34.33	17.10	0.85	0.23	3.81
RemoteCLIP	ResNet-50	−23.47	−9.12	−7.19	−11.49	−9.58	−0.40	−3.70	−4.04
	ViT-B/32	−26.09	−12.47	−13.23	−6.38	−13.57	−5.33	0.88	−2.32
SkyCLIP	ViT-L/14	−23.25	−13.68	−7.25	−19.29	−10.01	0.47	−1.00	−2.28
	ViT-B/32	−19.63	−20.15	−16.67	−12.77	−15.32	2.42	−0.80	−1.76
GeoRSCLIP	ViT-L/14	−32.09	−23.97	−14.14	−22.48	−12.27	1.75	0.16	−1.94
	ViT-B/32	−39.53	−34.17	−26.68	−17.87	−18.32	−7.80	−0.09	−4.96
GeoRSCLIP	ViT-L/14	−42.75	−31.52	−18.05	−27.18	−14.29	0.93	−0.04	−2.97
	ViT-H/14	−30.10	−27.97	−16.30	−21.10	−13.68	−0.18	−0.90	−3.09
Git-RSCLIP	ViT-B/16	−16.57	−8.72	−8.16	1.11	−9.06	−1.25	−1.29	−2.08
	ViT-L/16	−10.86	−7.72	2.09	−3.94	−0.98	−0.40	0.42	−0.34
PANEL (ours)	ViT-B/16	65.42	62.99	59.27	52.06	46.79	54.15	49.34	51.51

5.3.2. Semantic localization

We compute the cosine similarity between text and image features to generate activation maps for semantic localization. It is important to note that PANEL utilizes a binary prompting strategy during training (i.e., distinguishing only between PV-present and PV-absent). Consequently, while it does not support absolute semantic parsing of all land cover types, it is specifically optimized to highlight PV-relevant regions within complex scenes.

(1) *Quantitative evaluation.* Table 5 presents the quantitative results on the PV-SL benchmark using the mSC metric. (1) PANEL achieves an overall mSC of 51.51%, outperforming the second-best method, CLIP Surgery (5.06%), by a substantial margin of 46.45%. Notable gains are observed in both sub-meter and low-resolution subsets, confirming PANEL's cross-resolution robustness; (2) Models based on consistent v - v - v attention (CLIP Surgery and PANEL) yield positive mSC values that align well with visual perception. In contrast, standard CLIP models relying on q - k - v attention often produce negative responses; (3) When evaluating absolute mSC values, generalist models like RemoteCLIP, SkyCLIP, and Git-RSCLIP perform significantly worse than the original CLIP, with only GeoRSCLIP showing comparable results. This quantitative failure further demonstrates that existing RS CLIP models were likely pre-trained without sufficient PV-related image-text pairs, preventing them from learning discriminative PV features.

(2) *Explainability analysis of vision-language alignment.* We conduct a qualitative comparison among the top-performing models: CLIP Surgery (ViT-B/16, overall mSC = 5.06%), GeoRSCLIP (ViT-B/32, overall mSC = −4.96%), and our proposed PANEL (ViT-B/16, overall mSC =

51.51%). Notably, GeoRSCLIP demonstrates a strong suppression tendency in PV regions. To facilitate intuitive visual comparison, we invert GeoRSCLIP's similarity maps (subtracting values from 1), ensuring that higher scores consistently correspond to PV-relevant activations across all methods.

Fig. 7 presents representative results across diverse scenarios, including tiny rooftop installations, large clustered solar farms, and challenging environments (desert, mountain, farmland). Compared to CLIP Surgery and GeoRSCLIP, PANEL consistently produces more accurate and spatially coherent activations, with red-highlighted regions closely matching the true PV locations. Compared to CLIP Surgery and GeoRSCLIP, PANEL consistently produces more accurate and spatially coherent activations, with red-highlighted regions closely matching the true PV locations. The success of PANEL in accurately highlighting PV panels at the pixel level demonstrates that our binary prompting strategy effectively leverages the VLM paradigm to establish robust spatial correspondences.

Furthermore, the visualization results validate the effectiveness of our proposed binary prompt strategy. Although PANEL was trained using only simple “PV-present” vs. “PV-absent” descriptions, it successfully generalizes to identify PV panels across varied backgrounds. However, a limitation is observed in the scenario “Ground & Cluster & Obscured by clouds” (last column of Fig. 7). This suggests that while binary prompts are sufficient for general identification, they struggle to disentangle PV features from strong interferences like cloud occlusion. This finding highlights the necessity of introducing fine-grained, multi-class semantic descriptions (e.g., “PV panels obscured by clouds”) in future iterations to maintain localization precision in highly complex, obscured scenarios.

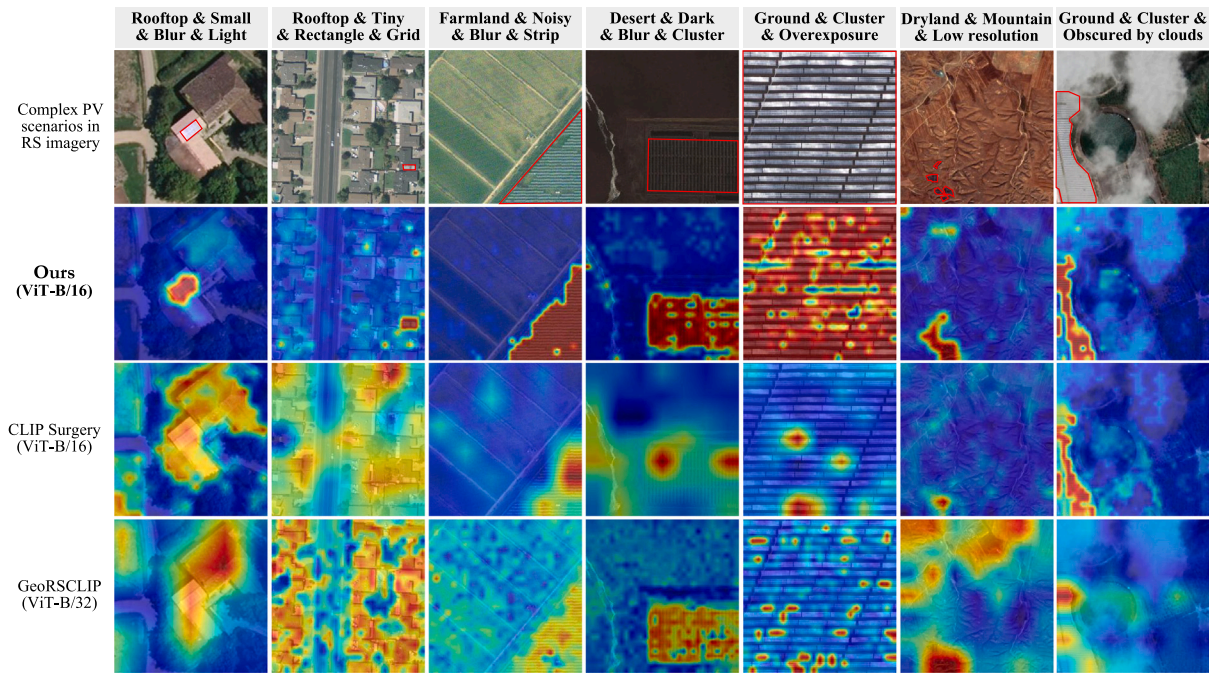


Fig. 7. Qualitative results of vision-language alignment for complex PV scenarios using our proposed PANEL (ViT-B/16), CLIP Surgery (ViT-B/16), and GeoRSCLIP (ViT-B/32). The red regions indicate higher confidence in the PV panels, while the blue regions indicate lower confidence.

Table 6

Zero-shot PV segmentation and object detection results of PANEL and representative RS CLIP models on the PV-SegA, PV-SegB, PV-SegC, and PV-Det benchmarks, evaluated by IoU, precision, recall, F1-score (%) at the threshold of 0.5.

Method	Version	PV-SegA			PV-SegB			PV-SegC			PV-Det		
		IoU	Recall	F1-score	IoU	Recall	F1-score	IoU	Recall	F1-score	Precision	Recall	F1-score
CLIP	ViT-B/16	18.25	32.21	30.87	22.57	37.18	36.82	3.74	44.06	7.22	14.17	21.39	17.05
CLIP Surgery	ViT-B/16	32.52	48.35	49.08	32.60	48.99	49.17	6.45	38.96	12.11	30.11	36.28	32.91
RemoteCLIP	ResNet-50	17.13	44.23	29.25	14.09	29.94	24.70	3.98	21.6	7.66	7.12	3.24	4.45
SkyCLIP	ViT-L/14	36.78	51.81	53.78	31.13	47.69	47.48	10.13	35.66	18.4	12.07	35.94	18.07
GeoRSCLIP	ViT-B/32	25.43	45.46	40.54	27.79	45.15	43.49	5.29	50.35	10.05	19.73	9.59	12.9
Git-RSCLIP	ViT-B/16	26.81	78.94	42.28	22.50	62.72	36.74	3.74	55.91	7.21	5.12	6.72	5.81
PANEL(ours)	ViT-B/16	55.18	84.95	71.12	61.31	87.81	76.01	12.95	94.71	22.92	24.24	60.10	34.55

5.3.3. Segmentation and object detection

To quantitatively evaluate the zero-shot segmentation and object detection capabilities of PANEL and other representative CLIP models, we derive predictions by binarizing the spatial similarity maps produced by semantic localization. Each pixel whose similarity value exceeds a given threshold is classified as PV panel while the remaining pixels are assigned to the background. For models with inverse activation (e.g., GeoRSCLIP), the similarity map is inverted prior to thresholding. The resulting binary masks serve as segmentation predictions, from which the minimum bounding rectangles are further extracted for detection evaluation.

Table 6 summarizes segmentation (IoU, recall, F1-score) and detection (precision, recall, F1-score) results at a threshold of 0.5. PANEL significantly outperforms all baselines across PV-SegA, PV-SegB, PV-SegC, and PV-Det. On PV-SegA (0.1–0.3 m), PANEL achieves 55.18% IoU and 71.12% F1-score, surpassing the second-best SkyCLIP by +18.40% IoU. The advantage is more pronounced on PV-SegB (0.6–1.0 m), where PANEL achieves 61.31% IoU—nearly doubling CLIP Surgery (32.60%). On PV-SegC (10–20 m), PANEL maintains the highest IoU (12.95%) and recall (94.71%), demonstrating robust cross-resolution generalization.

To provide evaluation beyond a single threshold, we generate precision–recall (PR) curves by varying the binarization threshold from 0.1 to 0.9 on PV-SegA/B/C. As shown in Fig. 8, PANEL consistently dominates the upper-right region of the PR space across all three benchmarks. Specifically, on PV-SegA, PANEL maintains precision above 0.8

up to recall ≈ 0.6 , while competing methods drop below 0.6. This margin widens on PV-SegB, where PANEL maintains precision near 0.8 at recall 0.6 while all other methods fall below 0.5. Even on PV-SegC, where the coarse 10–20 m resolution degrades all methods, PANEL still achieves the largest area under the PR curve, confirming its robustness across resolution bands.

For object detection on PV-Det, PANEL achieves the highest F1-score (34.55%) and substantially improves recall from 36.28% (CLIP Surgery) to 60.10%, demonstrating its strong zero-shot inference. However, its precision (24.24%) is lower than CLIP Surgery (30.11%). This precision–recall trade-off arises from two factors: (1) converting dense activation maps into bounding boxes via minimum bounding rectangles inherently introduces more candidate regions, and (2) the binary prompt strategy produces broad activations that can merge adjacent panels or activate PV-similar structures, thereby boosting recall at the expense of instance-level precision.

5.3.4. Image generation with stable diffusion

To assess the zero-shot generative capabilities of PANEL and other representative CLIP models, we conduct a background inpainting task. For each model, the binary mask generated for the segmentation and detection tasks is obtained by applying a similarity threshold of 0.5 to isolate the PV regions. The frozen inpainting model is then employed to regenerate the background, using the respective mask from each CLIP

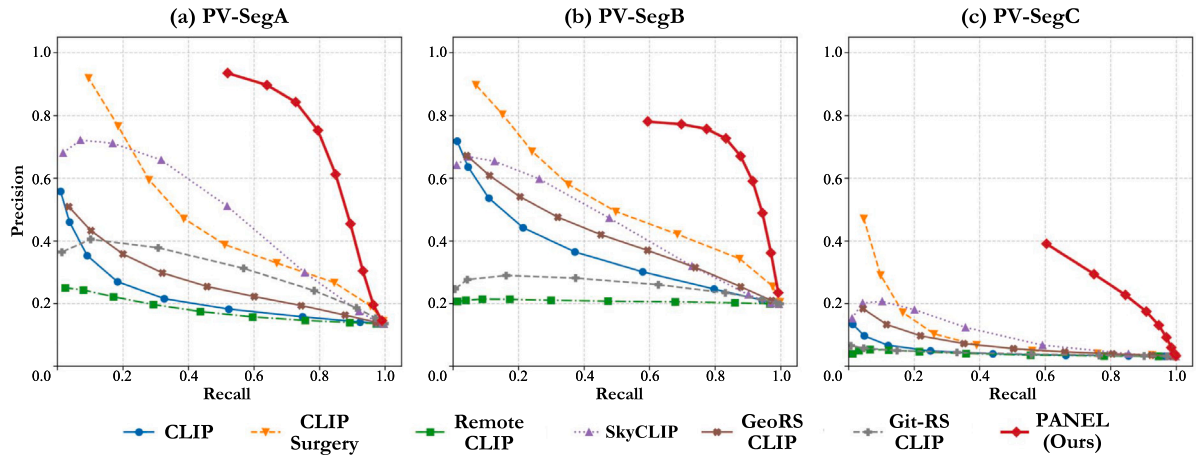


Fig. 8. Precision–recall curves for zero-shot PV segmentation on (a) PV-SegA (0.1–0.3 m), (b) PV-SegB (0.6–1.0 m), and (c) PV-SegC (10–20 m), obtained by varying the binarization threshold from 0.1 to 0.9. PANEL consistently occupies the upper-right region across all resolution bands, indicating higher precision at equivalent recall levels.

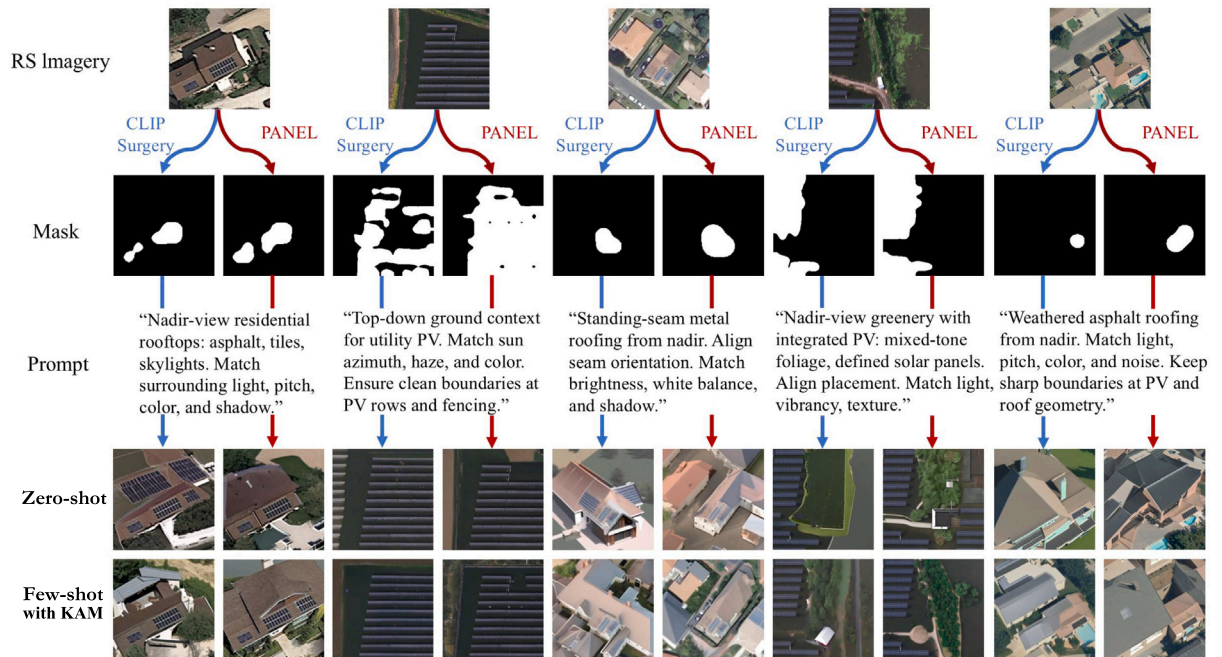


Fig. 9. Qualitative Stable Diffusion results conditioned on CLIP Surgery vs. our PANEL, comparing zero-shot performance and few-shot learning with KAM.

model. All background generations are performed with a CFG scale of $w = 7.5$.

The qualitative results across five specific generative scenarios – residential rooftops, utility ground context, standing-seam metal roofing, integrated greenery, and weathered asphalt roofing – are illustrated in Fig. 9. Visually, our PANEL approach demonstrates a clear advantage over CLIP Surgery in maintaining the structural coherence of PV panels. While CLIP Surgery occasionally introduces artifacts or blurred boundaries at the panel-background interface, PANEL produces sharper edges and more realistic background textures that align better with the original PV geometry. This suggests that our representations provide more stable semantic-geometric constraints during the generative process.

The quantitative results are summarized in Fig. 10(a), where performance is measured by the CLIP score delta. Here, Δ_{CLIP} represents the overall score difference, while $\Delta_{CLIP-1} - \Delta_{CLIP-5}$ correspond to the

five scenarios mentioned above. PANEL achieves the lowest overall Delta score (7.64%), surpassing all competing methods. This result indicates a superior ability to maintain the structural integrity and appearance of PV panels during the background modification process. Compared to the next-best models, CLIP Surgery (7.83%) and SkyCLIP (7.89%), PANEL demonstrates a clear and consistent advantage, while other models like Git-RSCLIP (8.32%) show significantly higher deviation. This robustness is further confirmed at the prompt-level, where PANEL achieves the best scores on four out of the five distinct prompts ($\Delta_{CLIP-1/2/3/5}$), highlighting the stability of our representations in guiding generation while preserving key semantic features. CLIP Surgery shows a slight lead on Δ_{CLIP-4} ; this is likely due to its refined spatial attention, which excels at processing the mixed-tone foliage and high-vibrancy textures specified in the greenery-integrated prompt.

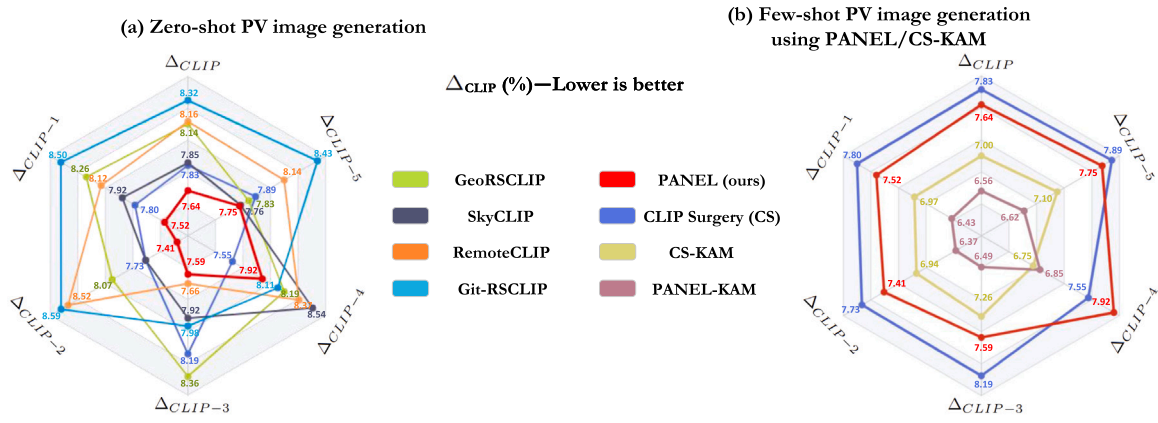


Fig. 10. Comparison of PV image generation performance under (a) zero-shot and (b) few-shot settings. Performance is evaluated by the overall (Δ_{CLIP}) and per-prompt (Δ_{CLIP-1} to Δ_{CLIP-5}) scores.

Table 7

Few-shot segmentation results of PANEL/CLIP Surgery(CS)-KAM module combined with four baselines on the PV-SegA (train: 25, val: 48,937), PV-SegB (train: 5, val: 6029), and PV-SegC (train: 250, val: 507,206) benchmarks, evaluated by IoU (%) and F1-score at the threshold of 0.5.

Method	KAM type	PV-SegA		PV-SegB		PV-SegC	
		IoU	F1-score	IoU	F1-score	IoU	F1-score
DeepLabV3+	–	81.26	89.66	51.69	65.16	46.32	65.20
	CS	+1.68	+0.95	+10.36	+8.69	+7.87	+5.09
	Ours	+3.25	+1.64	+19.17	+14.90	+14.91	+13.46
OCRNet	–	80.24	87.95	48.85	62.59	45.16	64.04
	CS	+1.88	+1.04	+11.32	+10.51	+10.04	+7.63
	Ours	+3.70	+1.82	+20.74	+16.19	+15.73	+14.08
SegFormer	–	80.74	88.48	51.50	66.03	45.41	64.35
	CS	+1.90	+1.06	+10.52	+8.00	+8.24	+5.54
	Ours	+3.49	+1.51	+19.35	+14.03	+15.67	+13.96
MaskFormer	–	82.05	89.94	49.67	64.89	45.52	64.69
	CS	+1.26	+0.91	+11.38	+8.80	+8.63	+5.42
	Ours	+2.82	+1.59	+20.17	+14.35	+16.26	+14.32

5.4. Analysis of few-shot learning using PANEL-KAM

In this subsection, we analyze PANEL and the proposed KAM module under few-shot learning settings across all PV-related tasks, focusing on cross-task and cross-baseline generalization and ablation studies.

5.4.1. Few-shot performance across tasks and baselines

(1) *Semantic segmentation.* We evaluate PANEL-KAM and CS-KAM (CLIP Surgery-based KAM) combined with four segmentation baselines: DeepLabV3+, OCRNet, SegFormer, and MaskFormer. Experiments are conducted under few-shot settings on PV-SegA, PV-SegB, and PV-SegC, using only 25, 5, and 250 training samples respectively, corresponding to approximately 0.05%, 0.08%, and 0.05% of the full datasets.

Table 7 summarizes the performance of these baselines with and without KAM integration. Both KAM variants significantly improve IoU and F1-score across all benchmarks, while PANEL-KAM consistently outperforms CS-KAM, with IoU gains of approximately 1.5–3.5% on PV-SegA, 11%–20% on PV-SegB, and 9%–15% on PV-SegC. Compared to PV-SegA, which comprises high-resolution imagery (0.1–0.3 m) of clearly visible distributed PV panels, the improvements on PV-SegB are more pronounced due to lower resolution and fewer training samples. Notably, on PV-SegC, which includes 507,206 validation images at 10–20 m resolution, both KAM modules still exhibit strong cross-resolution generalization with substantial gains in both metrics.

The left five columns of Fig. 11 present qualitative segmentation results using DeepLabV3+. The 1st column shows that PANEL-KAM yields higher recall and fewer false positives than CS-KAM, achieving an IoU

Table 8

Few-shot object detection results of our PANEL-based and CLIP Surgery(CS)-based KAM module combined with three baselines on the PV-Det (train: 20, val: 2522) benchmark, evaluated by AP@0.5/0.75/0.5:0.95 (%) at IoU threshold of 0.5 and multiple IoU thresholds from 0.5 to 0.95 in steps of 0.05.

Method	KAM type	PV-Det		
		AP@0.5	AP@0.75	AP@0.5:0.95
Faster R-CNN	–	13.4	1.0	3.9
	CS	+18.8	+10.2	+8.5
	Ours	+21.4	+11.9	+9.6
YOLOX	–	20.0	2.6	7.7
	CS	+17.6	+8.5	+7.4
	Ours	+3	+10.3	+8.8
DINO	–	30.9	15.7	15.9
	CS	+12.3	+6.2	+5.5
	Ours	+16.7	+7.9	+7.5

of 82%. Under few-shot training, the baseline alone fails to detect dark, texture-blurred PV panels (2nd column), whereas both KAM variants successfully recover these missed regions by leveraging CLIP/PANEL-derived priors. The 3rd to 5th columns further confirm that PANEL-KAM maintains clear advantages on lower-resolution imagery. However, KAM integration can occasionally introduce false positives: the yellow circle in the 1st column highlights a region where CS-KAM

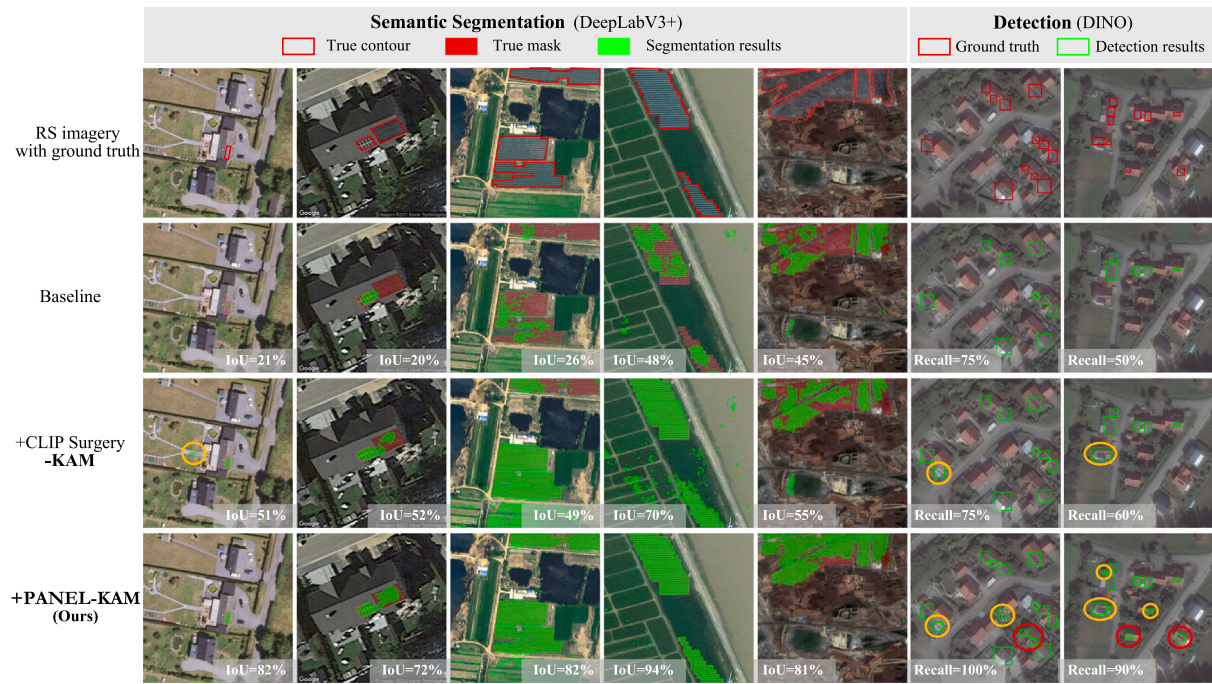


Fig. 11. Qualitative few-shot segmentation (DeepLabV3+) and detection (DINO) results with and without CLIP Surgery-KAM (CS-KAM) and PANEL-KAM. Yellow circles highlight false positives introduced by KAM-injected priors on spectrally similar non-PV regions; red circles indicate small-scale or spectrally ambiguous PV panels that only PANEL-KAM successfully detects.

misidentifies some ground regions as PV, indicating that injected priors may activate on spectrally similar non-PV regions.

(2) *Object detection.* We evaluate three detection baselines – Faster R-CNN, YOLOX, and DINO – with and without the PANEL/CS-KAM module on PV-Det, using 20 training and 2522 validation samples.

Table 8 and the rightmost two columns of Fig. 11 present the quantitative and qualitative results. Both KAM variants yield substantial AP improvements; in particular, PANEL-KAM raises DINO’s AP@0.5 from 30.9% to 47.6%. While KAM significantly boosts recall and discovers ambiguous targets, the qualitative results also reveal specific failure cases inherent to the binary prompting paradigm. As indicated by the yellow circles in Fig. 11, the injected priors can occasionally introduce false positives on spectrally similar non-PV structures, such as skylights and metallic rooftops. Because the binary priors provide broad category-level activations rather than fine-grained instance boundaries, PANEL-KAM produces slightly more of these false positive failure cases than CS-KAM due to its stronger overall PV-specific sensitivity. Nevertheless, the red circles highlight the complementary advantage: PANEL-KAM successfully detects small-scale and spectrally ambiguous PV panels that both the baseline and CS-KAM miss, confirming that PANEL-derived priors provide effective guidance for challenging instances.

(3) *Image generation.* We extended our few-shot analysis to the PV image generation, focusing on the task of inpainting. As shown in Fig. 9, PANEL-KAM produces backgrounds with significantly higher visual fidelity and tighter structural adherence to the textual prompts. For instance, our method generates sharper boundaries for utility PV rows and more realistic roof textures compared to CS-KAM. This qualitative performance is further supported by our quantitative metrics. We quantify the comparison using the Δ_{CLIP} metric, with results presented in Fig. 10(b). Notably, PANEL-KAM attains a superior overall score of 6.56%, validating its efficacy in few-shot scenarios. While CS-KAM holds a marginal lead on the fourth prompt involving complex greenery textures, PANEL-KAM demonstrates more robust generalization across the remaining scenarios, successfully matching environmental lighting and geometric constraints while strictly preserving the PV regions.

5.4.2. Ablation study of KAM

To clarify why KAM is necessary beyond PANEL pre-training alone, we conduct four groups of ablation experiments under the same few-shot settings described above, using DeepLabV3+ for segmentation (PV-SegA/B/C) and DINO for detection (PV-Det). The default configuration uses the full prompt library (1116 positive prompts), inserts KAM into all encoder layers of the baseline, and activates both the gated convolutional layer (G) and cross-attention fusion layer (C).

Specifically, we ablate: (1) *Component contribution:* the individual contribution of G and C by evaluating Baseline, Baseline+G, Baseline+C, and Baseline+KAM(G+C); (2) *Prompt sensitivity:* the effect of prompt diversity by varying the number of positive PV-relevant prompts (100%, 75%, 50%, 25% of the full library); (3) *Adaptation strategy:* the necessity of vision-language priors by comparing KAM against a vision-only adaptation strategy that directly connects PANEL’s image encoder to the baseline decoder head; and (4) *Insertion position:* the effect of KAM placement at different positions in the baseline architecture. In the encoder setting (default), KAM is inserted after the last layer of each encoder stage. The decoder setting follows the same principle: KAM is appended after the last layer of each decoder stage when multiple stages exist, or immediately before the task-specific prediction head otherwise. In the neck setting, a single KAM module is inserted as an intermediate layer between the encoder and decoder.

(1) *Component contribution.* As shown in row 1 of Fig. 12, both G and C consistently improve the baseline, and their combination yields the largest gains. For DeepLabV3+, the combined G+C raises IoU by +3.25% on PV-SegA, +19.17% on PV-SegB, and +14.91% on PV-SegC, with particularly pronounced effects at lower resolutions and tighter label budgets. For DINO, G+C achieves AP@0.5 of 47.5% (+16.7%), confirming that both prior injection and cross-attention alignment are essential for effective few-shot adaptation.

(2) *Prompt sensitivity.* Row 2 of Fig. 12 shows that KAM performance degrades gracefully as the prompt count is reduced. At 75%, results remain close to the full-library setting across all benchmarks (e.g., +3.16% vs. +3.25% IoU on PV-SegA; +17.58% vs. +19.17% on PV-SegB). However, reducing to 25% leads to noticeable drops, particularly on PV-SegC (from +14.91% to +4.21% IoU) and PV-Det (from

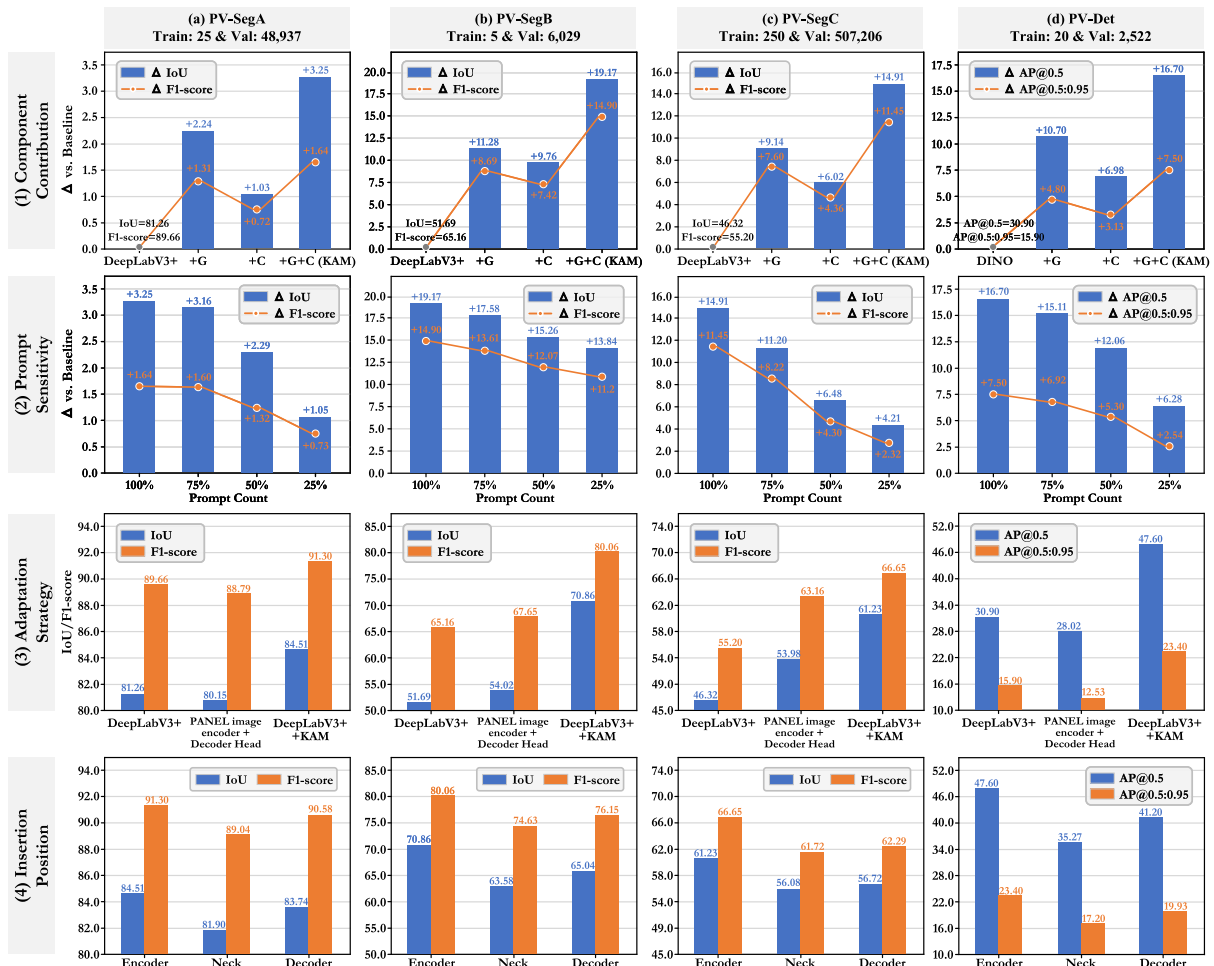


Fig. 12. Ablation study of KAM on few-shot segmentation (DeepLabV3+, columns a-c) and detection (DiNO, column d). Row 1: component contribution (G: gated convolutional layer; C: cross-attention fusion layer). Row 2: prompt sensitivity (percentage of the full 1116-template library). Row 3: adaptation strategy comparison between vision-only transfer (PANEL image encoder + decoder head) and our full KAM pipeline. Row 4: KAM insertion position (encoder, neck, or decoder).

+16.70% to +6.28% AP@0.5). These results validate our ensemble prompting strategy: a sufficiently diverse library stabilizes the averaged text features for similarity computation, thereby improving the quality of spatial priors injected into KAM.

(3) *Adaptation strategy.* Row 3 of Fig. 12 compares the full KAM pipeline (Baseline+KAM) against a vision-only strategy that directly connects PANEL’s frozen image encoder to the baseline decoder head (PANEL image encoder+Head), bypassing all text-derived priors. On PV-SegA, PANEL image encoder+Head (IoU=80.15%) underperforms the baseline (81.26%), and a similar regression occurs on PV-Det (AP@0.5: 28.02% vs. 30.90%). This degradation stems from the single-scale nature of the PANEL image encoder output (16×16 patches), which cannot capture diverse PV panel sizes across multi-source benchmarks, especially where multi-scale feature pyramids are critical (Lin et al., 2017). On the more homogeneous PV-SegB and PV-SegC, PANEL image encoder+Head does improve over the baseline (e.g., 54.02% vs. 51.69% IoU on PV-SegB), yet remains substantially below KAM (70.86%). These results confirm that the few-shot gains of KAM cannot be replicated by a simple vision-only encoder-head transfer, and that the text-derived spatial priors provide essential complementary guidance.

(4) *Insertion position.* Row 4 of Fig. 12 shows that encoder insertion consistently achieves the best performance. Neck insertion, providing only a single interaction point, yields the lowest gains (e.g., IoU=63.58% on PV-SegB vs. 70.86% for encoder), confirming that one-time injection is insufficient. Decoder insertion outperforms neck

placement but still underperforms encoder insertion. This is because PANEL’s patch-level priors are inherently coarse, and injecting them at the decoder stage – where representations are progressively refined for task-specific prediction – can introduce spatial misalignment (Ronneberger et al., 2015). In contrast, encoder insertion allows the baseline to integrate and refine these coarse priors through its full hierarchical pipeline before task-specific decoding.

5.5. Label-efficient solution on sample selection

Annotating large-scale RS datasets remains labor-intensive and costly, so selecting a small yet informative subset from a massive unlabeled pool is crucial for practical deployment. Beyond sample quantity, sample diversity is a key driver of downstream generalization: prioritizing diverse samples yields greater gains than random selection of the same size.

Specifically, we leverage VLM-derived similarity scores to guide diversity-aware sample selection. For each unlabeled image, we compute its global similarity score with the PV-present positive prompts using either the original CLIP or our proposed PANEL. Based on these 1D similarity scores, we compare three distinct clustering-based selection strategies: (1) *DBSCAN*, which identifies density-based clusters and draws samples uniformly across them; (2) *k-means*, which partitions the score distribution into k groups and samples proportionally from each; and (3) *k-center greedy* (Sener and Savarese, 2017), which iteratively

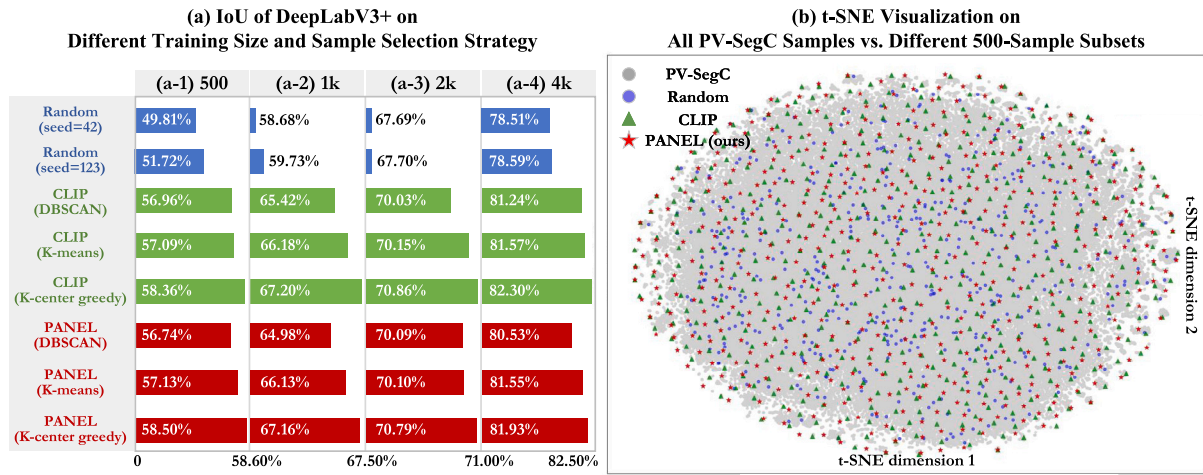


Fig. 13. Evaluation of label-efficient sample selection strategies on the PV-SegC benchmark. (a) Downstream segmentation performance (IoU of DeepLabV3+) across varying sample budgets (500, 1k, 2k, and 4k) using random selection (with two independent seeds) and VLM-guided selection (CLIP and PANEL) clustered via DBSCAN, *k*-means, and *k*-center greedy. (b) t-SNE visualization of ResNet-50 embeddings, comparing the full PV-SegC dataset (gray points) against 500-sample subsets selected by random sampling with the seed is 42 (blue circles) and the *k*-center greedy strategy guided by CLIP (green triangles) and PANEL (red stars). Notably, while PANEL provides uniform core coverage, CLIP's selections reach further into the sparse outer boundaries of the feature space, reflecting its broader semantic variance.

maximizes coverage by first selecting a random sample as the initial center, then repeatedly choosing the sample whose minimum distance to all existing centers is the largest, until the desired budget is reached.

To rigorously evaluate these sample selection strategies, we used the PV-SegC benchmark, which provides the most comprehensive pool (507,456 images) to assess selection robustness. As a comparative baseline, we perform the most common random sampling using two independent seeds (42 and 123), thereby eliminating potential biases from specific random initializations. For all strategies, we evaluate performance across varying sample budgets of 500, 1k, 2k, and 4k images from the PV-SegC dataset. At each budget, the selected subset is used to train the DeepLabV3+ model, while the remaining unselected images serve as the test set to verify the robustness of our approach.

Fig. 13(a) presents the quantitative segmentation performance across the different sample budgets and selection strategies. (1) Both VLM-guided approaches (CLIP and PANEL) consistently outperform random selection across all three clustering methods, confirming that VLM-based similarity scores effectively serve as a reliable proxy for guiding sample selection. This advantage is particularly pronounced under extremely tight budgets: at 500 samples, random selection yields an IoU of only 49.81% to 51.72%, whereas the VLM-guided strategies boost performance significantly, with PANEL reaching up to 58.50%; (2) Among the clustering techniques, the *k*-center greedy consistently yields the best performance, followed by *k*-means and then DBSCAN. This superiority stems from the explicit distance-maximization mechanism of *k*-center greedy, which ensures the broadest possible coverage of the sample space; (3) For the optimal *k*-center greedy strategy, PANEL slightly outperforms CLIP at the tight 500-sample budget (58.50% vs. 58.36%), but CLIP maintains a consistent lead at larger budgets (e.g., 82.30% vs. 81.93% at 4k). This highlights a fundamental trade-off: PANEL's binary pre-training tightly clusters PV-positive samples, which benefits zero-shot interpretation but compresses the feature space, reducing the variance needed for diversity-based sampling. Conversely, CLIP's open-vocabulary pre-training captures subtle contextual differences, yielding a broader semantic space. Consequently, given a sufficient sample budget, CLIP's generalist representations provide a superior basis for maximizing sample diversity.

Fig. 13(b) visualizes the selected samples using t-SNE applied to ResNet-50 embeddings. The gray points represent the entire feature space of the 507,456 PV-SegC images, while the colored points denote the 500-sample subsets. Although the data points are densely

overlapped, the overall distribution patterns of the selection strategies remain distinctly observable. Random selection (blue circles) exhibits noticeable local aggregation, leaving large regions of the feature space unsampled. In contrast, both VLM-guided strategies yield significantly more dispersed selections that better span the global distribution. Crucially, while PANEL (red stars) achieves excellent uniformity within the dense core of the PV distribution, CLIP (green triangles) pushes further into the extreme edges and sparse boundaries of the feature space. This visual evidence corroborates our finding: CLIP's fully utilized text modality effectively pulls image features apart, creating a more expansive and varied semantic space that is highly advantageous for diversity-based sampling.

6. Conclusion

This study presents PANEL, a PV-specific VLM that unifies zero-shot and few-shot PV interpretation tasks in RS. We curate the first worldwide PV vision-language dataset comprising over one million image-text pairs spanning 0.1–20 m resolutions, paired with a structured binary prompt library that enables robust vision-language alignment without dense captions. Building on CLIP (ViT-B/16), PANEL is retrained end-to-end with bi-directional InfoNCE, comprehensive augmentations, and randomized prompt diversification. To extract PV-relevant priors from the frozen PANEL, we introduce PANEL Surgery, which augments the image encoder with a parallel *v-v-v* attention branch to produce global similarity scores, spatial similarity maps, and patch-level features that directly support zero-shot inference. For few-shot adaptation, we design a plug-and-play KAM module that injects these priors into diverse baseline architectures via gated convolution and cross-attention fusion, thereby bridging PANEL's frozen semantic priors with the task-specific learning capacity required by downstream models under limited supervision.

To enable rigorous evaluation, we establish seven PV-specific benchmarks spanning five interpretation tasks. PANEL consistently surpasses representative CLIP models in zero-shot classification, semantic localization, segmentation, detection, and PV-preserving image generation. In few-shot settings, KAM generalizes across four segmentation architectures and three detection frameworks, delivering substantial gains with minimal labels. Extensive ablation studies further confirm the complementary roles of each KAM component, validate the necessity of vision-language priors over vision-only transfer, and establish encoder

insertion as the optimal integration strategy. Additionally, we propose a label-efficient sampling strategy that leverages VLM-derived similarity scores with DBSCAN, k -means, and k -center greedy clustering, consistently outperforming random selection under tight annotation budgets. Collectively, these contributions establish PANEL as a PV-specific VLM for robust and data-efficient interpretation in real-world RS applications for solar energy.

For future work, we plan to expand the PV vision-language dataset beyond binary prompts to fine-grained, multilingual descriptions enriched with land-use contexts and geospatial metadata, and to incorporate additional modalities such as multispectral, thermal, and meteorological data. These advances will support the development of a PV-specific large language model with instruction-following and multi-modal reasoning capabilities, enabling a broader suite of applications from array topology parsing to power forecasting and planning support.

CRedit authorship contribution statement

Ruizhe Deng: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis. **Zhiling Guo:** Writing – review & editing, Supervision, Methodology. **Penglei Zhang:** Validation, Software, Investigation, Data curation. **Jiaze Li:** Writing – original draft, Validation, Software, Methodology. **Xin Xu:** Methodology. **Qi Chen:** Conceptualization. **Yuntian Chen:** Writing – review & editing, Supervision, Funding acquisition. **Jinyue Yan:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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